First Week

System

$x(t) \rightarrow \boxed{S} \rightarrow y(t)$

- Single input single output (SISO)
- Multiple input multiple output (MIMO)

$x_1(t) \rightarrow$, $x_2(t) \rightarrow \boxed{S} \rightarrow y_1(t)$, $\rightarrow y_2(t)$

LINEAR System:

$x_1(t) \rightarrow \boxed{S} \rightarrow y_1(t)$

$x_2(t) \rightarrow \boxed{S} \rightarrow y_2(t)$

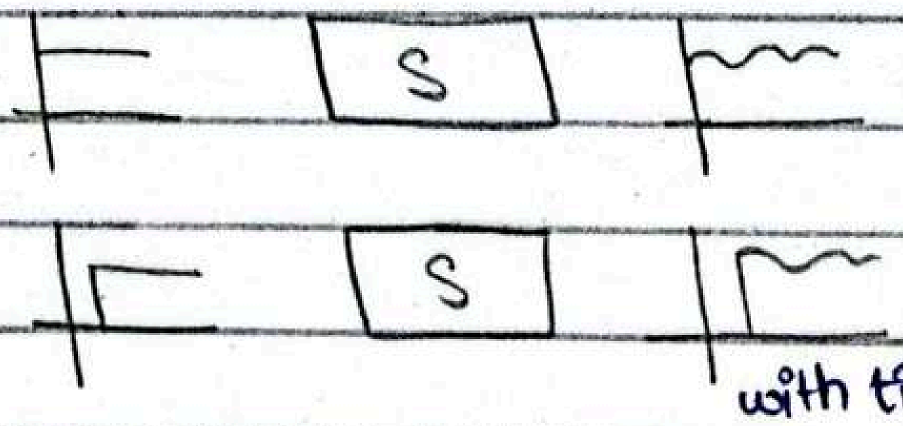
Properties:

$ax_1(t)+bx_2(t) \rightarrow \boxed{S} \rightarrow y_1(t)+y_2(t)$

↳ addition

$ax_1(t) \rightarrow \boxed{S} \rightarrow ay_1(t)$

↳ homogenity



} Time invariant

with time delay

* no feedback system — open loop

Components of a Control System:

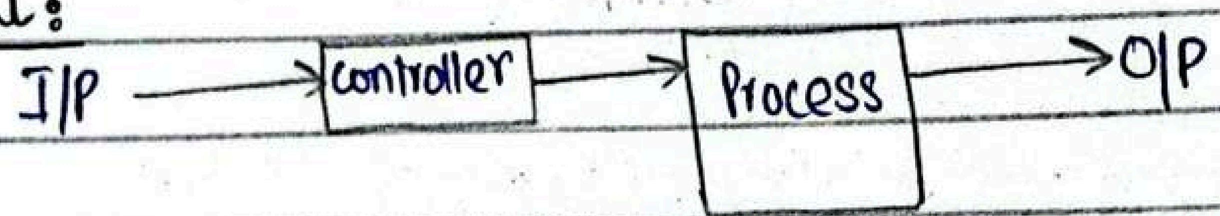
- Input
- Output
- ways to achieve input & output
- control action

Basic Technologies

- **Plant | Process:** portion of the system which is to be controlled or regulated.
- **Controller:** controls the plant | process

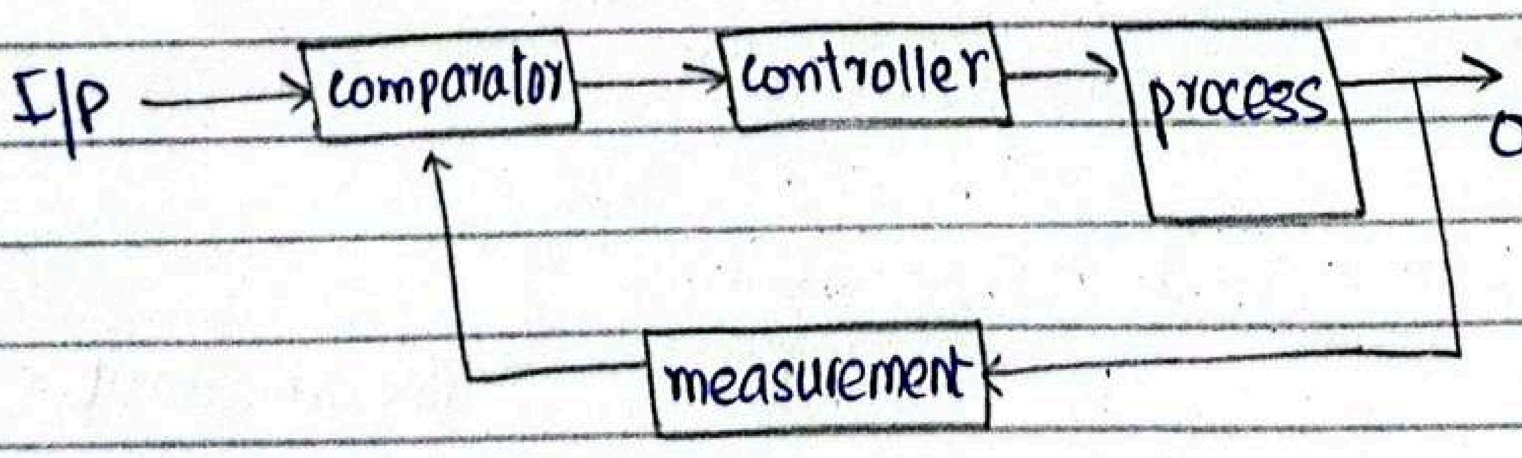
- input
- output
- disturbance

Input:

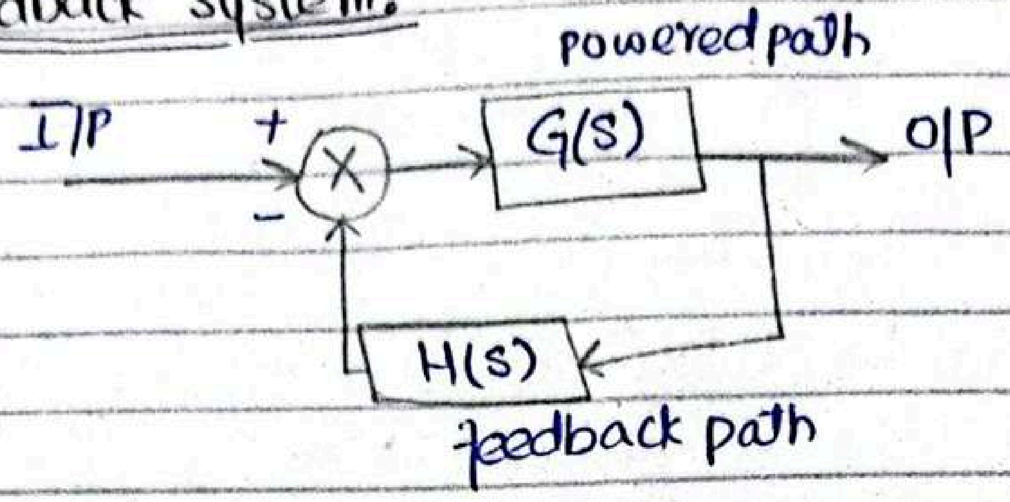


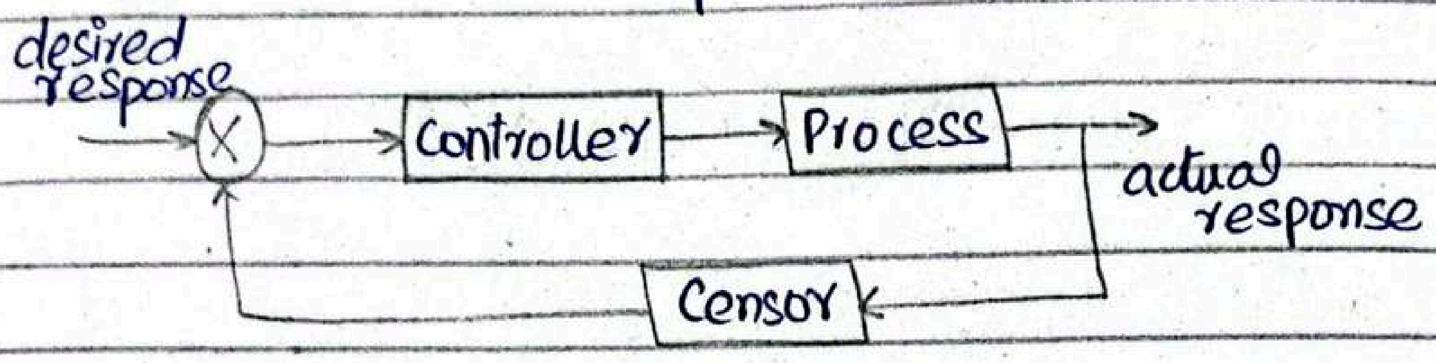
Open loop: O/P is neither measured nor feedback.

Closed Loop Control System



Feedback System:





Chapter no 2

$$x(t) \longrightarrow \boxed{S} \longrightarrow y(t)$$

Transfer function : $T.F = \dfrac{Y(s)}{X(s)}$

Laplace transform : (covers a breeder range of syste

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$

Fourier transform : (for signal analysis)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

* fourier is a subset of Laplace transform

Review of Laplace Transforms

$\delta(t) \rightarrow \boxed{S} \rightarrow h(t)$

any i/p wave

$h(t) \xleftrightarrow{\mathcal{L}} H(s)$

$h(t) \xleftrightarrow{F} H(j\omega)$

$\Rightarrow$ if $s = \sigma + j\omega$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt = \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}\,dt$$

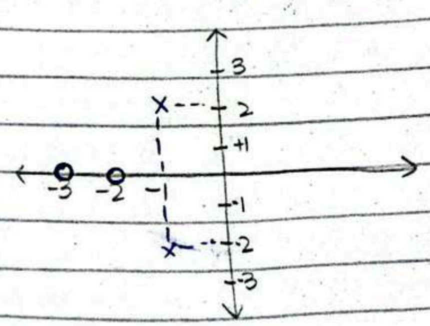
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt \quad \text{(fourier)}$$

Example 1 $\quad H_1(s) = \dfrac{(s+2)(s+3)}{(s+1+2j)(s+1-2j)}$

zeros $\Rightarrow s=-2, s=-3$; poles $\Rightarrow s = -1-2j, s=-1+2j$

zeros — represented by 'O'

poles — represented by 'X'

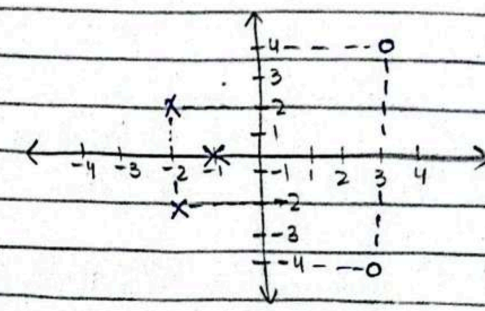


Eg#2

$$H_1(s) = \frac{(s-3+4j)(s-3-4j)}{(s+1)(s+2+2j)(s+2-2j)}$$

$s-3+4j=0$; $s-3-4j=0$; $s=-1$; $s+2+2j=0$, $s+2-2j=0$

$s = 3-4j \quad s=3+4j \quad s=-2-2j \quad s=-2+2j$



Impulse related properties

1) $x_1(t) = \delta(t)$

$$X_1(s) = \int_{-\infty}^{\infty} \delta(t)e^{-st}\,dt = e^{-s(0)} = 1$$

2) $x_2(t) = u(t)$

$$X_2(s) = \int_{-\infty}^{\infty} u(t)e^{-st}\,dt$$

$$= \int_0^{\infty} e^{-st}\,dt = \frac{e^{-st}}{-s}\Big|_0^{\infty} = -\frac{1}{s}\left(e^{-\infty}-e^{0}\right) = \frac{1}{s}$$

3) $x_3(t) = tu(t)$

$$X_3(s) = \int_{-\infty}^{\infty} tu(t)e^{-st}\,dt = \int_0^{\infty} te^{-st}\,dt$$

$$= \frac{te^{-st}}{-s}\Big|_0^{\infty} - \int_0^{\infty} 1\cdot\frac{e^{-st}}{-s}\,dt = \frac{1}{s}\,\frac{e^{-st}}{-s}\Big|_0^{\infty} = \frac{1}{s^2}$$

4) $x_4(t) = e^{-at}u(t)$

$$X_4(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt$$

$$= \int_0^{\infty} e^{-(s+a)t}dt = \frac{e^{-(s+a)t}}{-(s+a)}\Bigg|_0^{\infty} = \frac{1}{s+a}$$

5) $x_5(t) = \sin\omega t \, u(t)$ $\qquad \because \sin\omega t = \dfrac{e^{j\omega t} - e^{-j\omega t}}{2j}$

$$= \frac{e^{-j\omega t} - e^{-j\omega t}}{2j}u(t) \qquad \because \cos\omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2j}$$

$$x_5(s) = \frac{1}{2j}\int_0^{\infty} e^{j\omega t}e^{-st}dt - \frac{1}{2j}\int_0^{\infty} e^{-j\omega t}e^{-st}dt$$

$$= \frac{1}{2j}\int_0^{\infty} e^{-(s-j\omega)t}dt - \frac{1}{2j}\int_0^{\infty} e^{-(s+j\omega)t}dt$$

$$= \frac{1}{2j}\left[\frac{e^{-(s-j\omega)t}}{-(s-j\omega)}\Bigg|_0^{\infty} - \frac{e^{-(s+j\omega)}}{-(s+j\omega)}\Bigg|_0^{\infty}\right]$$

$$X_5(s) = \frac{1}{2j}\left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega}\right] = \frac{1}{2j}\left[\frac{\cancel{s}+j\omega-\cancel{s}+j\omega}{s^2+\omega^2}\right] = \frac{\omega}{s^2+\omega^2}$$

Properties of Laplace Transform

1) Linearity:

$$ax_1(t) + bx_2(t) \longleftrightarrow ax_1(s) + bx_2(s)$$

2) Time-Shifting:

$$x(t-t_0) \xleftrightarrow{\mathcal{L}} X(s)e^{-st_0}$$

3) Frequency Shifting:

$$e^{-at}x(t) \longleftrightarrow X(s+a)$$

4) Differentiation property:

$$\frac{dx(t)}{dt} \longleftrightarrow sX(s) - x(0)$$

$$\frac{dx^2(t)}{dt^2} \longleftrightarrow s^2X(s) - sx(0) - x'(0)$$

$$\frac{dx^n(t)}{dt^n} \longleftrightarrow s^nX(s) - \sum_{k=1}^{n} s^{n-k}x^{k-1}(0)$$

5) Integration property:

$$\int_0^t x(\tau)d\tau \longleftrightarrow \frac{X(s)}{s}$$

Initial value theorem

$$f(0^+) = \lim_{s\to\infty} sF(s)$$

Final value theorem

$$f(\infty) = \lim_{s\to 0} s(F(s))$$

→ 3 types of analysis

1- Transient 2- Steady state 3- stability

→ $\dfrac{d^2y(t)}{dt^2} + 12\dfrac{dy(t)}{dt} + 32y(t) = 32x(t)$

if initial condition are zero

taking laplace transform

$$s^2Y(s) + 12sY(s) + 32Y(s) = 32X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{32}{s^2+12s+32}$$

Let say $x(t)=u(t)$, we want to find output

$$X(s) = u(s) = \frac{1}{s}$$

$$Y(s) = \frac{32}{s(s^2+12s+32)} = \frac{32}{s(s+4)(s+8)}$$

$$= \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+8}$$

$$32 = A(s+4)(s+8) + Bs(s+8) + Cs(s+4)$$

$s=0$

$32 = A(0+4)(0+8)$, $s=-8$, $s=-4$

$A=1$; $B=-2$; $C=1$

$$Y(s) = \frac{1}{s} - \frac{2}{s+4} + \frac{1}{s+8}$$

→ Now taking inverse Laplace transform

$$y(t) = u(t) - 2e^{-4t}u(t) + e^{-8t}u(t)$$
$$= u(t)\left[1-2e^{-4t}+e^{-8t}\right]$$

→ $F(s) = \dfrac{2}{(s+1)(s+2)^2}$

$$= \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

$$2 = A(s+2)^2 + B(s+1)(s+2)^2 + C(s+1)$$

$s=-1$, $s=-2$

$$2 = A(-1+2)^2 + B(-1+1)(-1+2)^2 + C(-1+1)$$
$$2 = A(1)^2 \Rightarrow \boxed{A=2}$$

$$2 = A(-2+2)^2 + B(-2+1)(-2+2)^2 + C(-2+1)$$
$$2 = C(-1) \Rightarrow \boxed{C=-2}$$

$$4A + 2B + C = 2$$
$$4(2) + 2B + (-2) = 2$$
$$\boxed{B=-2}$$

$$F(s) = \frac{2}{s+1} - \frac{2}{s+2} - \frac{2}{(s+2)^2}$$

$$f(t) = 2e^{-t}u(t) - 2e^{-2t}u(t) - 2e^{-2t}tu(t)$$
$$= 2u(t)\left[e^{-t} - e^{-2t} - te^{-2t}\right]$$

→ $F(s) = \dfrac{3}{s(s^2+2s+5)}$

$$= \frac{A}{s} + \frac{Bs+C}{s^2+2s+5}$$

$$3 = A(s^2+2s+5) + Bs+C(s)$$

$s=0$

$\boxed{A = \frac{5}{3}}$ Now,

$A+B=0$; $2A+C=0$

$B = -\frac{3}{5}$; $C = -\frac{6}{5}$

$$F(s) = \frac{3/5}{s} - \frac{3/5\,s + 6/5}{s^2+2s+5}$$

$$= \frac{3/5}{s} - \frac{3}{5}\,\frac{s+2}{s^2+2s+5} = \frac{3/5}{s} - \frac{3}{5}\,\frac{s+2}{s^2+2s+1+4}$$

$$= \frac{3/5}{s} - \frac{3}{5}\,\frac{s+2}{(s+1)^2+(2)^2}$$

$$= \frac{3/5}{s} - \frac{3}{5}\left[\frac{s+1}{(s+1)^2+2^2} + \frac{1}{(s+1)^2+2^2}\right]$$

$$F(s) = \frac{3/5}{s} - \frac{3}{5}\left[\frac{s+1}{(s+1)^2+2^2} + \frac{1}{2}\,\frac{2}{(s+1)^2+2^2}\right]$$

$$f(t) = \frac{3}{5}u(t) - \frac{3}{5}\left[e^{-t}\cos 2t + \frac{1}{2}e^{-t}\sin 2t\right]u(t)$$

$$= \left\{\frac{3}{5} - \frac{3}{5}e^{-t}\left[\frac{\cos 2t + \frac{1}{2}\sin 2t}{\sqrt{1+\left(\frac{1}{2}\right)^2}}\right]u(t)\right\}$$

$$= 0.6 - 0.67e^{-t}\left[\frac{1}{\sqrt{1+(1/2)^2}}\cos 2t + \frac{1/2}{\sqrt{1+(1/2)^2}}\sin 2t\right]$$

$\therefore \cos(A-B) = \cos A\cos B + \sin A \sin B$

$$= 0.6u(t) - 0.67e^{-t}\cos(2t-\varphi)\,u(t)$$

$$\varphi = \tan^{-1}\frac{1}{2}$$

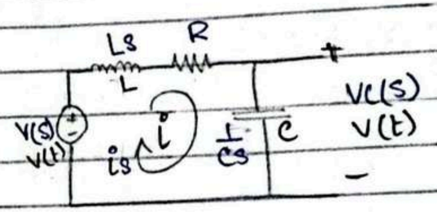
$$G(s) = \frac{2s+1}{s^2+2s+6}$$

$$(s^2+2s+6)\,Y(s) = (2s+1)\,X(s)$$

$$s^2Y(s) + 2sY(s) + 6Y(s) = 2sX(s) + X(s)$$

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 6y(t) = 2\frac{dx(t)}{dt} + x(t)$$

Transfer function of Electrical Networks:



$\rightarrow i = C\frac{dV_c}{dt}$

$I(s) = Cs\,V(s)$

$\rightarrow V_L = L\frac{di_L}{dt}$

$V_L(s) = LsI(s)$

Find $\frac{V_c(s)}{V(s)}$

$$-v(t) + L\frac{di}{dt} + Ri + V_c(t) = 0$$

$$-v(t) + LC\frac{d^2V_c(t)}{dt^2} + RC\frac{dV_c}{dt} + V_c(t) = 0$$

taking Laplace transform

$$LCs^2V_c(s) + RCs\,V_c(s) + V_c(s) + V(s)$$

$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}$$

$$= \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

→ in terms of 's'

$$-V(S) + LSI(S) + RI(S) + I(S)\frac{1}{Cs} = 0$$

$$\left(Ls + R + \frac{1}{Cs}\right)(I(s)) = V(S)$$

$$\left(Ls + R + \frac{1}{Cs}\right)(Cs)\,(V_C(S)) = V(s)$$

$$(LCs^2 + RCs + 1)(V_C(S)) = V(s)$$

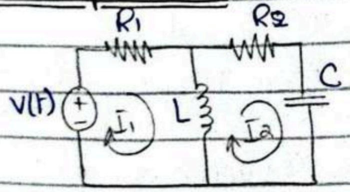
$$\frac{V_C(S)}{V(S)} = \frac{1}{LCs^2 + RCs + 1}$$

→ 3rd Method

Apply VDR:

$$V_C(s) = V(s)\,\frac{1/Cs}{1/Cs + R + Ls}$$

<u>Multiple Loop Circuit</u>



$$-V(S) + R_1 I_1(S) + LS\,(I_1(S) - I_2(S)) = 0$$

$$R_2 I_2(S) + \frac{1}{Cs} I_2(s) + Ls\,(I_2(S) - I_1(S)) = 0$$

$$(Ls + R_1)I_1(S) - LsI_2(S) = V(S)$$

$$-LsI_1(S) + \left(\frac{1}{Cs} + (Ls + R_2)I_2(S)\right) = 0$$

$$\begin{bmatrix} Ls + R_1 & -Ls \\ -Ls & \frac{1}{Cs} + Ls + R_2 \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$

$$I_1(S) = \frac{\begin{bmatrix} V(S) & -Ls \\ 0 & R_2 + Ls + \frac{1}{Cs} \end{bmatrix}}{\Delta}$$

$$I_2(S) = \frac{\begin{bmatrix} Ls + R_1 & (V(s)) \\ -Ls & 0 \end{bmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} Ls + R_1 & -Ls \\ -Ls & R_2 + Ls + \frac{1}{Cs} \end{vmatrix}$$

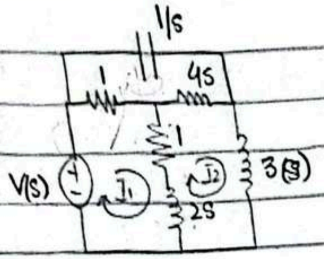
$$= (Ls + R_1)\left(R_2 + \frac{1}{Cs} + Ls\right) - L^2s^2$$

<u>Shortcut Method:</u>

$$\begin{bmatrix} \text{Sum of imp} \\ \text{around Mesh 1} \end{bmatrix} I_1(s) - \begin{bmatrix} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{bmatrix} I_2(S) = \begin{bmatrix} \text{sum of} \\ \text{voltage in 1} \end{bmatrix}$$

$$\begin{bmatrix} \text{sum of imp} \\ \text{around Mesh 2} \end{bmatrix} I_2(S) - \begin{bmatrix} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{bmatrix} I_1(S) = \begin{bmatrix} \text{sum of} \\ \text{voltage in 2} \end{bmatrix}$$

(4)



Find $\frac{I_1(s)}{V(s)}$

Mesh 1:

$$(2+2s)(I_1(s)) - (1+2s)(I_2(s)) - (1)(I_3(s)) = V_s \quad -(1)$$

Mesh 2:

$$(1+9s)(I_2(s)) - (1+2s)(I_1(s)) - 4s(I_3(s)) = 0 \quad -(2)$$

Mesh 3:

$$\left(1+4s+\frac{1}{s}\right)(I_3(s)) - (1)(I_1(s)) - 4s(I_2(s)) = 0 \quad -(3)$$

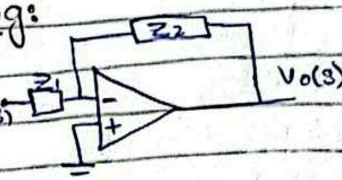
$$\begin{bmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \end{bmatrix}$$

$$I_1(s) = \frac{\begin{bmatrix} V_s & -(1+2s) & -1 \\ 0 & (9s+1) & -4s \\ 0 & -4s & \frac{1}{s}+4s+1 \end{bmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{vmatrix}$$

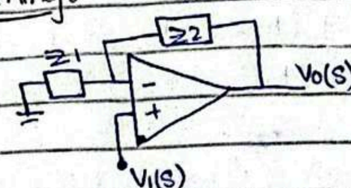
Transfer function of Active ccts/OP-AMP based ccts

Inverting:



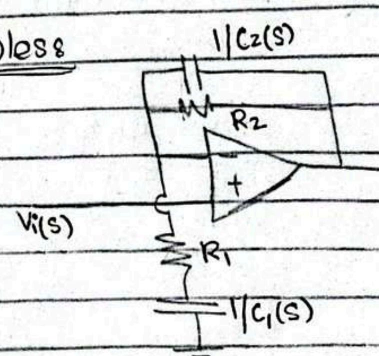
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

Non-Inverting:



$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1}$$

Examples:



$$Z_2 = R_2 \parallel \frac{1}{C_2(s)} = \frac{R_2 \; {}^1/C_2 s}{R_2 + \frac{1}{C_2(s)}}$$

$$= \frac{R_2}{1+R_2C_2(s)} \quad -(1)$$

$$Z_1 = R_1 + \frac{1}{C_1(s)} = \frac{1+R_1C_1(s)}{C_1(s)} \quad -(2)$$

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{R_2 / 1+R_2C_2(s)}{\frac{1+R_1C_1(s)}{C_1(s)}}$$

$$= 1 + \frac{R_2C_1(s)}{(1+R_2C_2(s))(1+R_1C_1(s))}$$

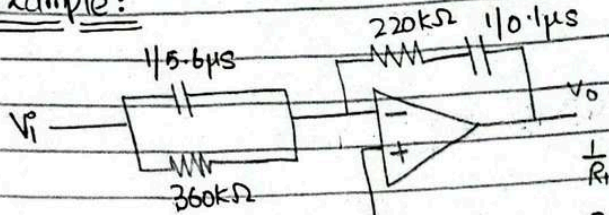
$$= \frac{1 + (R_1C_1S)(1+R_2C_2(S)) + R_2C_1S}{(1+R_1C_1S)(1+R_2C_2S)}$$

$$= \frac{R_1C_1R_2C_2s^2 + (R_1C_1 + R_2C_2 + R_2C_1)s + 1}{R_1C_1R_2C_2s^2 + (R_1C_1 + R_2C_2)s + 1}$$

∴ Poles = Power of s in denominator

∴ Zeros = Power of s in numerator

Example:



$$\frac{1}{R_1} + \frac{1}{L} = \frac{L+R_1}{LR_1}$$

$$R_1 + L = R_1 + L$$

$$Z_2 = 220k + \frac{10^7}{s}, \quad Z_1 = \frac{360k}{1+(360k)(5.6\mu)s}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{220k + 10^7/s}{360k / 1+(360k)(5.6\mu)s}$$

$$= -\left[\frac{220ks + 10^7}{s} \times \left(\frac{(1+2.016)s}{360k}\right)\right]$$

$$= -\left(\frac{1.232s^2 + 56.61s + 27.77}{s}\right)$$

$$= -\left[1.232s + \frac{27.77}{s} + 56.61\right]$$

Transfer Function of Mechanical Systems

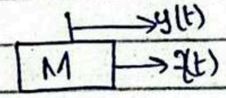
→ Translational Mechanical System → Mass, Spring, Damper
→ Force, Displacement

Mass:

→ stores kinetic energy

→ rigid bodies

∴ Number of equations depends on the point of displacement.

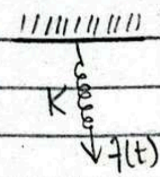


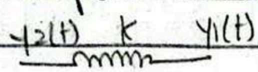
$f(t) \propto a$

$f(t) = Ma \Rightarrow \boxed{f(t) = M\frac{d^2y(t)}{dt}}$

2- **Spring:**

→ stores potential energy





$f(t) = k(y_1(t) - y_2(t))$

$f(t) \propto y(t)$

$\boxed{f(t) = ky(t)}$

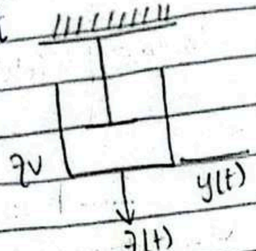
3- Damper:

→ Frictional element

$f(t) \propto v(t)$

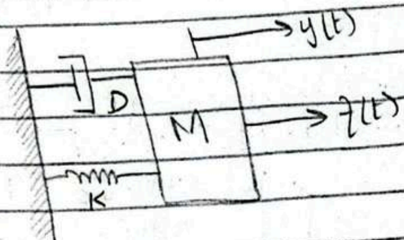
$f(t) \propto \frac{dy(t)}{dt}$

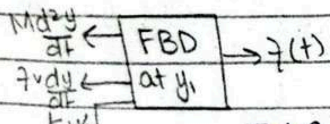
$f(t) = f_v \frac{dy(t)}{dt}$



→ f_v = viscous function coeffecient

Ideal Spring Mass Damper System





$$f(t) = M\frac{d^2y}{dt^2} + f_v\frac{dy}{dt} + ky(t)$$

Taking Laplace transform

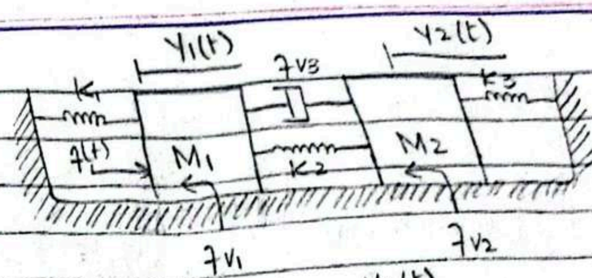
$$F(S) = Ms^2Y(S) + f_v s Y(S) + k(Y(S))$$

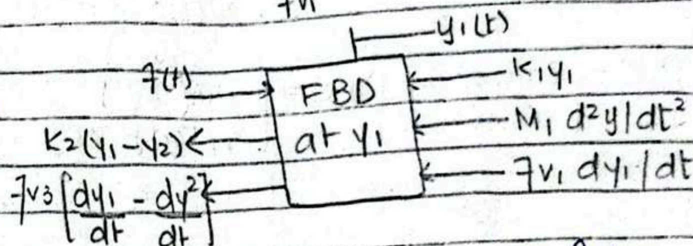
$$Y(S)[Ms^2 + f_v s + k] = F(S)$$

$$\frac{Y(S)}{F(S)} = \frac{1}{Ms^2 + f_v s + k}$$

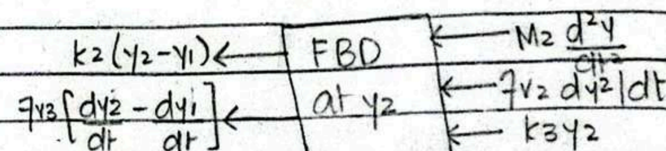
$M = 3,\ f_v = 2,\ k = 1$

$$\frac{Y(S)}{F(S)} = \frac{1}{3s^2 + 2s + 1}$$





$$f(t) = M_1\frac{d^2y_1}{dt^2} + f_{v1}\frac{dy_1}{dt} + f_{v3}\left[\frac{dy_1}{dt} - \frac{dy_2}{dt}\right] + k_1y_1 + k_2(y_1 - y_2)$$



$$M_2\frac{d^2y}{dt^2} + f_{v2}\frac{dy_2}{dt} + k_3y_2 + k_2(y_2 - y_1) + f_{v3}\left[\frac{dy_2}{dt} - \frac{dy_1}{dt}\right] = 0$$

$$\Rightarrow \left[M_1s^2 (f_{v1} + f_{v3})s + (k_1 + k_2)\right]Y_1(s) - (f_{v3}s + k_2)Y_2(s) = Fs$$

$$\Rightarrow -(f_{v3}s + k_2)Y_1(s) + \left[M_2s^2 + (f_{v2} - f_{v3})s + k_2 + k_3\right]Y_2(s) = 0$$

CRAMER'S RULE:

$$\begin{bmatrix} M_1s^2 + (f_{v1} + f_{v3})s + k_1 + k_2 & -f_{v3}s + k_2 \\ -(f_{v3}s + k_2) & M_1s^2 + (f_{v2} + f_{v3})s + k_2 + k_3 \end{bmatrix}\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} s^2+1 & -2 & -4s \\ -2 & 4s+2 & -5s \end{bmatrix}$$

Technique:

$$\begin{bmatrix}\text{sum of imp at} \\ \text{motion 1}\end{bmatrix} Y_1(s) - \begin{bmatrix}\text{sum of imp} \\ \text{common in 1 and 2}\end{bmatrix} Y_2(s) = \begin{bmatrix}\text{sum of} \\ \text{applied forces}\end{bmatrix}_{\text{at } Y_1}$$

$$\begin{bmatrix}\text{sum of imp at} \\ \text{motion 2}\end{bmatrix} Y_2(s) - \begin{bmatrix}\text{sum of imp} \\ \text{common}\end{bmatrix} Y_1(s) = \begin{bmatrix}\text{sum of} \\ \text{applied forces}\end{bmatrix}_{\text{at } Y_2}$$

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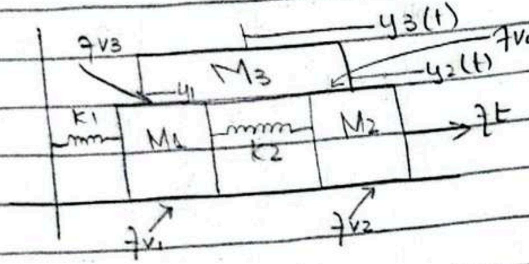
Example:



Equation by using technique

For M_1:

$$(M_1s^2 + (f_{v1}+f_{v3})s + k_1 + k_2)Y_1(s) - k_2Y_2(s) - (f_{v3}s)Y_3(s) = 0$$

For M_2:

$$(M_2s^2 + (f_{v2}+f_{v4})s + k_2)Y_2(s) - k_2Y_1(s) - (f_{v4}s)Y_3(s) = F(s)$$

For M_3:

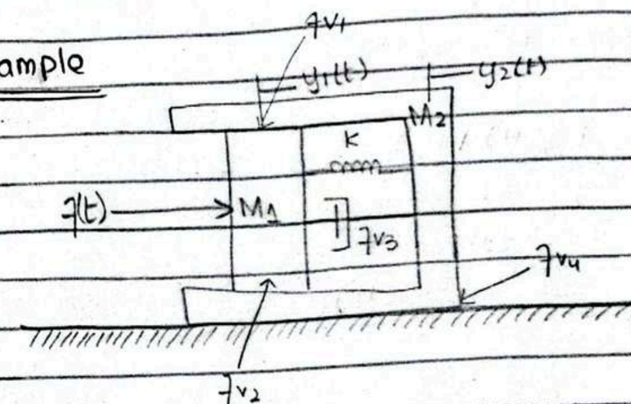
$$(M_3s^2 + (f_{v3}+f_{v4})s)Y_3(s) - (f_{v3}s)Y_1(s) - (f_{v4}s)Y_2(s) = 0$$

Equation using cramer's rule:

$$\begin{bmatrix} M_1s^2 + (f_{v1}+f_{v3})s + k_1 + k_2 & -k_2 & -f_{v3}s \\ -k_2 & M_2s^2 + (f_{v2}+f_{v4})s + k_2 & -f_{v4}s \\ -f_{v3}s & -f_{v4}s & M_3s^2 + (f_{v3}-f_{v4})s \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ Y_3) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \\ 0 \end{bmatrix}$$

Remember!
Matrix mein M, k, f ki value put karni hai

Example



Take $f_{v1} = f_{v2} = f_{v3} = f_{v4} = 1\ Ns/m$

$M_1 = M_2 = 1kg$

$k = 1N/m$

Find $\dfrac{Y_2(s)}{F(s)}$

For M_1:

$$(M_1s^2)Y_1(s) - (f_{v1}+f_{v2}+f_{v3}+k)Y_2(s) = F(s)$$

$$+(f_{v1}+f_{v2}+f_{v3})s + k)$$

$$(M_1s^2 + (f_{v1}+f_{v2}+f_{v3})s + k)Y_1(s)$$

For M_2:

$$\{M_2 s^2 + (f_{v1} + f_{v2} + f_{v3} + f_v)s + k\} - [(f_{v1} + f_{v2} + f_{v3})s + k]\, Y_1(s) = 0$$

Cramer's Rule

$$\begin{bmatrix} s^2+3s+1 & -(3s+1) \\ -(3s+1) & s^2+4s+1 \end{bmatrix} \begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$Y_2(s) = \frac{\begin{vmatrix} s^2+3s+1 & F(s) \\ -(3s+1) & 0 \end{vmatrix}}{\Delta}$$

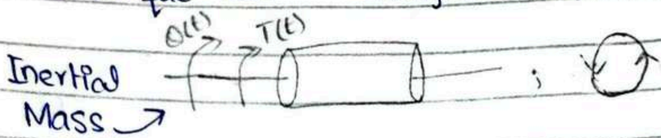
$$\Delta = s(s^3+7s^2+5s+1)$$

$$Y_2(s) = \frac{(3s+1)F(s)}{\Delta}$$

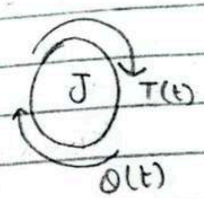
$$\frac{Y_2(s)}{F(s)} = \frac{3s+1}{s(s^3+7s^2+5s+1)}$$

Rotational Mechanical System

→ Torque $T(t)$ → Angular displacement $\theta(t)$

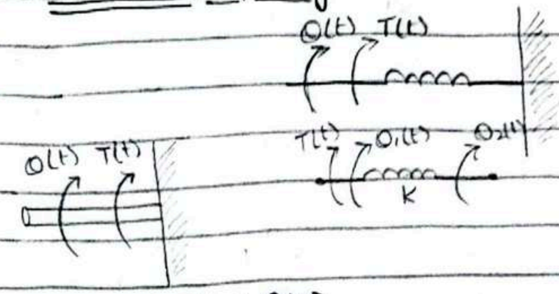


$$T(t) \propto \frac{d^2\theta(t)}{dt^2} \quad ; \quad T(t) = J\frac{d^2\theta(t)}{dt^2}$$



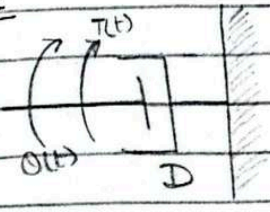
J = Moment of Inertia

→ **Torsional Spring:**



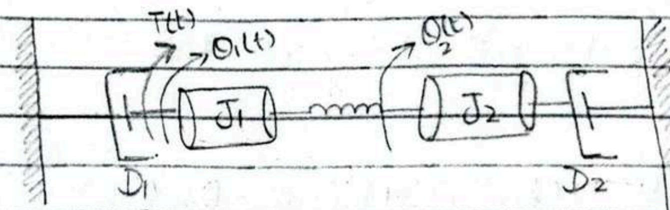
$$T(t) \propto \theta(t)$$
$$T(t) = k\theta(t)$$
$$T(t) = k(\theta_1(t) - \theta_2(t))$$

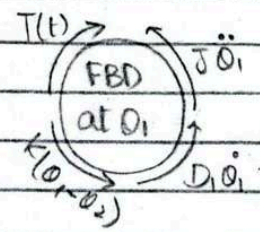
→ **Torsional damper:**



$$T(t) \propto \frac{d\theta(t)}{dt}$$
$$T(t) = D\frac{d\theta(t)}{dt}$$

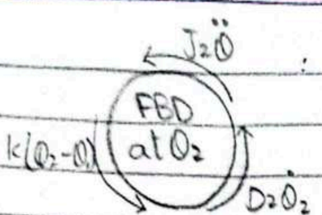
Example





$$J_1\frac{d^2\theta_1}{dt^2} + D_1\frac{d\theta_1}{dt} + k(\theta_1 - \theta_2) = T(t)$$

↳ $(J_1 s^2 + D_1 s + k)\theta_1(s) - k\theta_2(s) = T(s)$



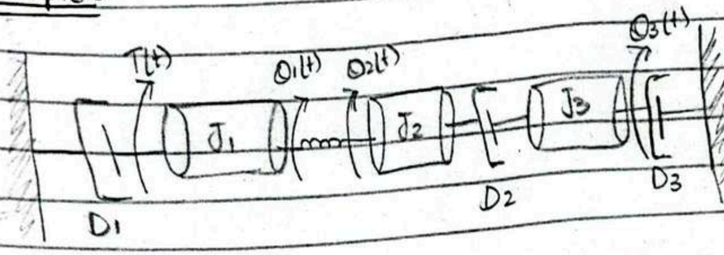
$$J_2\frac{d^2\theta_2}{dt} + D_2\frac{d\theta_2}{dt} + k(\theta_2-\theta_1) = 0$$

$$-k\theta_1(S) + (J_2S^2 + D_2S + k)\theta_2(S) = 0$$

Cramer's Rule:

$$\begin{bmatrix} J_1s^2+D_1s+k & -k \\ -k & J_2s^2+D_2s+k \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

Example:



For θ_1:

$$(J_1s^2 + D_1s + k_1)\theta_1(S) - k\theta_2(S) = T(S)$$

For θ_2:

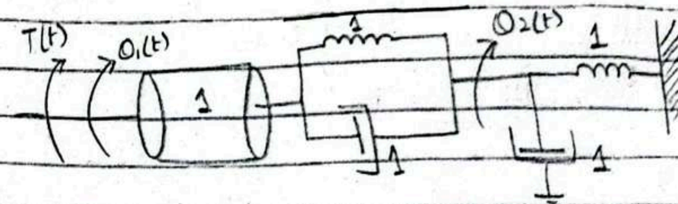
$$-k\theta_1(S) + (J_2S^2 + D_2S + k)\theta_2(S) - D_2S\,\theta_3(S) = 0$$

For θ_3:

$$-D_2S\theta_2(S) + (J_3S^2 + (D_2+D_3)S)\,\theta_3(S) = 0$$

Cramers' Rule:

$$\begin{bmatrix} J_1S^2+D_1(S)+k & -k & 0 \\ -k & J_2S+D_2S+k & -D_2S \\ 0 & -D_2S & J_3\theta_2+(D_2+D_3)S \end{bmatrix}\begin{bmatrix} \theta_1(S) \\ \theta_2(S) \\ \theta_3(S) \end{bmatrix} = \begin{bmatrix} T(S) \\ 0 \\ 0 \end{bmatrix}$$



Find $\dfrac{\theta_2(S)}{T(S)}$

$$(S^2+S+1)\theta_1(S) - (S+1)\theta_2(S) = T(S)$$

$$-(S+1)\theta_1(S) + (2S+2)\theta_2(S) = 0$$

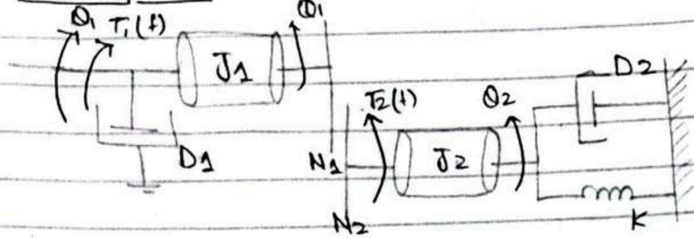
Cramer's Rule:

$$\begin{bmatrix} S^2+S+1 & -(S+1) \\ -(S+1) & 2S+2 \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(S) \\ 0 \end{bmatrix}$$

$$\theta_2(S) = \frac{\begin{vmatrix} S^2+S+1 & Ts \\ -(S+1) & 0 \end{vmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} S^2+S+1 & -(S+1) \\ -(S+1) & 2S+2 \end{vmatrix}$$

Example:



i) $\dfrac{\theta_2(s)}{T_2(s)} = ?$

$\rightarrow$ at θ_2 :

$$J_{eq} = J_2 + J_1\left(\frac{N_2}{N_1}\right)^2$$

$$D_{eq} = D_2 + D_1\left(\frac{N_2}{N_1}\right)^2$$

$$K_{eq} = K$$

$$\frac{\theta_2(s)}{T_2(s)} = \frac{1}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

ii) $\dfrac{\theta_1(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\theta_1(s) = T_1(s)$$

$$J_{eq} = J_1 + J_2\left(\frac{N_1}{N_2}\right)^2$$

$$D_{eq} = D_1 + D_2\left(\frac{N_1}{N_2}\right)^2$$

$$K_{eq} = K\left(\frac{N_1}{N_2}\right)^2$$

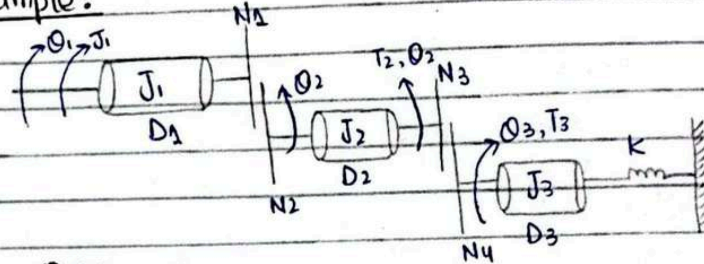
iii) $\dfrac{\theta_2(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\theta_2(s) = T_2(s)$$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\theta_2(s) = \frac{N_2}{N_1} \times T_1(s)$$

$$\frac{\theta_2(s)}{T_1(s)} = \frac{(N_2/N_1)}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

Example:



$\dfrac{\theta_3(s)}{T_3(s)} = ?$

$$J_{eq} = J_3 + J_2\left(\frac{N_4}{N_3}\right)^2 + J_1\left(\frac{N_4}{N_3}\right)^2\left(\frac{N_2}{N_1}\right)^2$$

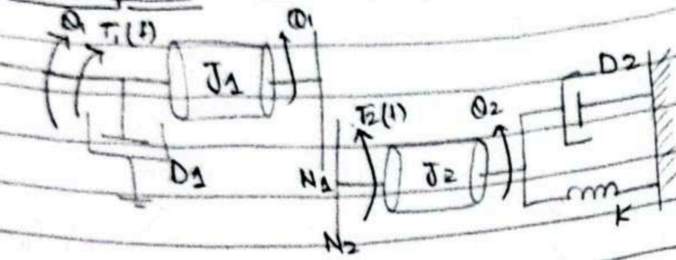
$$D_{eq} = D_3 + D_2\left(\frac{N_4}{N_3}\right)^2 + D_1\left(\frac{N_4}{N_3}\right)^2\left(\frac{N_2}{N_1}\right)^2$$

$$K_{eq} = K$$

Transfer FUNCTION of ElectroMechanical System (DC Servomotor)

- Servomotor is with feedback loop.

Example:



i) $\dfrac{\theta_2(s)}{T_2(s)} = ?$

→ at θ_2:

$$J_{eq} = J_2 + J_1\left(\frac{N_2}{N_1}\right)^2$$

$$D_{eq} = D_2 + D_1\left(\frac{N_2}{N_1}\right)^2$$

$$K_{eq} = K$$

$$\frac{\theta_2(s)}{T_2(s)} = \frac{1}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

ii) $\dfrac{\theta_1(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\,\theta_1(s) = T_1(s)$$

$$J_{eq} = J_1 + J_2\left(\frac{N_1}{N_2}\right)^2$$

$$D_{eq} = D_1 + D_2\left(\frac{N_1}{N_2}\right)^2$$

$$K_{eq} = K\left(\frac{N_1}{N_2}\right)^2$$

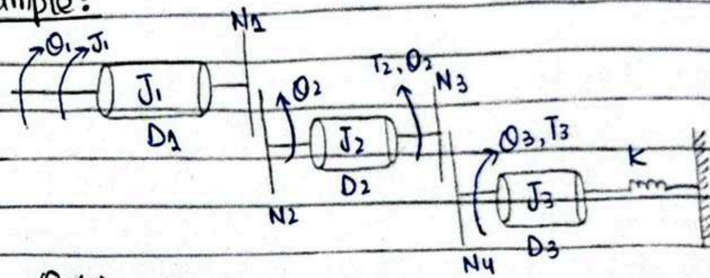
iii) $\dfrac{\theta_2(s)}{T_1(s)} = ?$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\,\theta_2(s) = T_2(s)$$

$$(J_{eq}s^2 + D_{eq}s + K_{eq})\,\theta_2(s) = \frac{N_2}{N_1} \times T_1(s)$$

$$\frac{\theta_2(s)}{T_1(s)} = \frac{(N_2/N_1)}{J_{eq}s^2 + D_{eq}s + K_{eq}}$$

Example:



$\dfrac{\theta_3(s)}{T_3(s)} = ?$

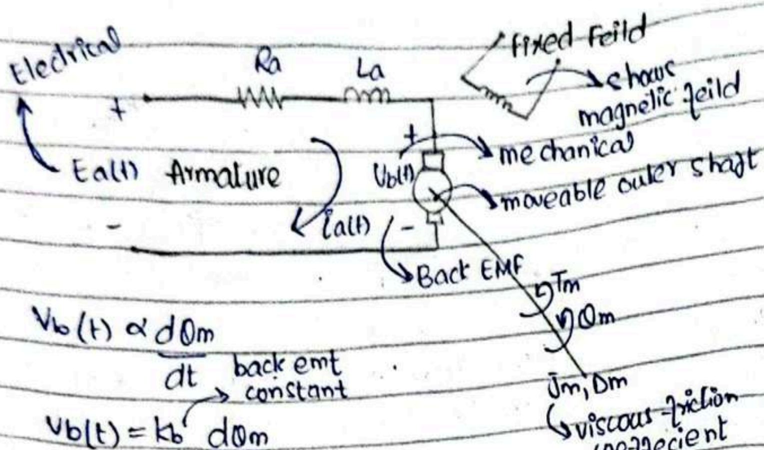
$$J_{eq} = J_3 + J_2\left(\frac{N_4}{N_3}\right)^2 + J_1\left(\frac{N_4}{N_3}\right)^2\left(\frac{N_2}{N_1}\right)^2$$

$$D_{eq} = D_3 + D_2\left(\frac{N_4}{N_3}\right)^2 + D_1\left(\frac{N_4}{N_3}\right)^2\left(\frac{N_2}{N_1}\right)^2$$

$$K_{eq} = K$$

Transfer Function of ElectroMechanical System (DC Servomotor)

- Servomotor is with feedback loop.



$V_b(t) \propto \frac{d\theta_m}{dt}$ back emf constant

$V_b(t) = K_b \frac{d\theta_m}{dt}$

$\Rightarrow V_b(s) = K_b s\ \theta_m(s)$ —— ①

Transfer Function:

$$\frac{\theta_m(s)}{E_a(s)}$$

$T_m(t) \propto i_a(t)$

$T_m(t) = K_t\ i_a(t) \Rightarrow T_m(s) = K_t I_a(s)$ —— (ii)

K_t → motor torque constant

$$-e_a(t) + R_a i_a(t) + L_a \frac{di_a}{dt} + V_b(t) = 0$$

$$-E_a(s) + R_a I_a(s) + L_a s I_a(s) + V_b(s) = 0$$

$$(R_a + L_a s) I_{as} + K_b s\ \theta_m(s) = E_{as} \quad \text{—— (iii)}$$

eq ①

→(ii) $I_a(s) = \frac{T_m(s)}{K_t}$

→(iii) $(R_a + L_a s)\frac{T_m(s)}{K_t} + K_b s\ \theta_m(s) = E_a(s)$ —— (iv)

$$J_m \frac{d^2\theta_m}{dt} + D_m \frac{d\theta_m}{dt} = T_m(t)$$

$$(J_m s^2 + D_m s)\,\theta_m(s) = T_m(s) \quad \text{—— ⑤}$$

→(iv) $(R_a + L_a s)\frac{(J_m s^2 + D_m s)}{K_t}\theta_m(s) + K_b s\,\theta_m(s) = E_a(s)$

$$\left[\frac{(R_a + L_a(s))(J_m s^2 + D_m(s)) + K_b(s)}{K_t}\right]\theta_m(s) = E_a(s)$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t}{(R_a + L_a(s))(J_m s^2 + D_m(s)) + K_b(s) K_t}$$

$L_a \to 0$

$$= \frac{K_t / R_a J_m}{s^2 + (D_m/J_m)s(K_b K_t / R_a J_m)s}$$

$$= \frac{K_t / R_a J_m}{s\left(s + 1/J_m\left(D_m + \frac{K_b K_t}{R_a}\right)\right)} = \frac{K}{s(s+a)}$$

$$= \frac{K}{s(s+a)}$$

$\therefore \alpha = s + \frac{1}{J_m}\left(D_m + \frac{K_b K_t}{R_a}\right)$

$\therefore K = \frac{K_t}{R_a J_m}$

Poles : $s = 0$, $s = -a$

Chapter no 03

State Space Representation

For non-Linear time equation

→ input , output
→ state variables

$\longrightarrow \boxed{\text{State } S} \longrightarrow$

$x_1(t)$, $x_2(t)$, $x_3(t)$, ...

$$\dot{X} = A\underbrace{X}_{\text{state vector}} + BU$$

$$Y = CX + D\underbrace{U}_{\text{input vector}}$$

A → state matrix

B → input matrix, C → output matrix

D → Direct transmission matrix

- How many state variables?
- How to assign state variables?
- Write state equations * 1st order } state space model
- Write output equations

$$\overset{n\times 1}{\dot{X}} = \overset{n\times n}{A}\underbrace{X}_{n\times 1} + \overset{n\times m}{B}\underbrace{U}_{m\times 1}$$

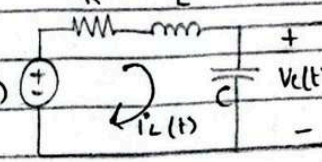
∴ P = output variables
n = state variables
m = input variables

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \ \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\overset{P\times 1}{Y} = \overset{P\times n}{C}\underbrace{X}_{n\times 1} + \overset{P\times m}{D}\underbrace{U}_{m\times 1}$$

∴ no of state variables = no. of state equations

Example



⇒ *energy storing variables = 2

⇒ state variables = 2

Let,

$x_1(t) = V_o(t) = v_o(t)$ —— ①

$x_2(t) = i_L(t)$

$i_L(t) = C\dfrac{dV_c(t)}{dt}$

$\Rightarrow x_2 = C\dfrac{dx_1}{dt} \Rightarrow \dfrac{dx_1}{dt} = \dfrac{1}{C} \times x_2$ —— ②

$-v(t) + Ri_L(t) + L\dfrac{di_L}{dt} + V_c(t) = 0$

$\dfrac{di_L}{dt} = -\dfrac{1}{L}V_c(t) - \dfrac{R}{L}i_L(t) + \dfrac{1}{L}v(t)$

$\dfrac{dx_2}{dt} = -\dfrac{1}{L}x_1 - \dfrac{R}{L}x_2 + \dfrac{1}{L}v(t)$ —— ③

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} v(t)$$

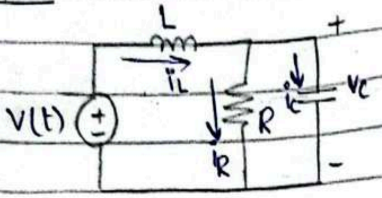
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Let, $V_R(t) = v_o$

$V_R(t) = Rx_2$

$$\begin{bmatrix} 0 & R \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example



Let,

$\because i_L = i_R + i_C$

$V_C = x_1 \quad ; \quad i_L = x_2$

$$i_C = C\frac{dV_C}{dt} \Rightarrow i_C = C\frac{dx_1}{dt} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C} i_C = \frac{1}{C}(i_L - i_R)$$

$$= \frac{1}{C}\left(i_L - \frac{V_C}{R}\right)$$

$$= \frac{1}{C}x_2 - \frac{1}{RC}x_1 \Rightarrow \frac{dx_1}{dt} = \frac{-1}{RC}x_1 + \frac{1}{C}x_2 \quad \text{— (i)}$$

$$V_L = L\frac{di_L}{dt} \Rightarrow \frac{dx_2}{dt} = \frac{1}{L}V_L \quad \text{— (ii)}$$

$$-V(t) + V_L + V_R = 0$$

$$V_L = V_t - V_C$$

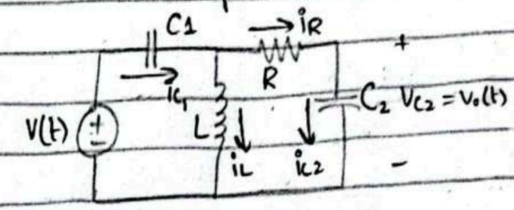
$$\frac{dx_2}{dt} = \frac{1}{L}\left(-V_C + v(t)\right)$$

$$\frac{dx_2}{dt} = -\frac{1}{L}x_1 + \frac{1}{L}v(t) \quad \text{— (iii)}$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \end{bmatrix} = \begin{bmatrix} -1/RC & 1/C \\ -1/L & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} V_b(t)$$

$$y = \begin{bmatrix} -1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v(t)$$

Obtain state space model. Take v_{C2} as output



Let,

$V_{C2} = x_3 = v_0(t)$ $\qquad x_1, x_2, x_3 \rightarrow$ state variables

$V_{C1} = x_1$

$$i_L = x_2 \Rightarrow \frac{dx_2}{dt} = C\frac{dV_{C2}(t)}{dt}$$

$$i_{C1} = C_1\frac{dV_{C1}}{dt} \Rightarrow \frac{1}{C_1}i_{C1} = \frac{d}{dt}V_{C1} \quad \text{—(i)}$$

$$i_{C2} = C_2\frac{dV_{C2}}{dt} \Rightarrow \frac{1}{C_2} \times i_{C2} = \frac{dV_{C2}}{dt} \quad \text{— (ii)}$$

$$V_L = L\frac{di_L}{dt} \Rightarrow \frac{1}{L} \times V_L = \frac{di_L}{dt} \quad \text{—(iii)}$$

Replace i_{C2} and V_L

$$-V(t) + V_{C1}(t) + V_L(t) = 0$$

$\rightarrow$ (iii) $-V(t) + V_{C1} + V_L = 0$

$$V_L = V(t) - V_{C1}$$

$\rightarrow$ (iii)

$$\frac{di_L}{dt} = \frac{1}{L}\left(V(t) - V_{C1}\right) \Rightarrow \frac{dx_2}{dt} = \frac{1}{L}\left(V(t)\right) - \frac{1}{L}x_1$$

$$i_{C1} = i_R + i_L$$

$$i_R = \frac{V_L - V_{C2}}{R}$$

→ (v) $i_{c1} = i_L + \left(\frac{V_L - V_{c2}}{R}\right)$

$$i_{c_1} = i_L + \frac{1}{R}\left[\underbrace{-V_{c1} + v(t)}_{V_L} - V_{c2}\right]$$

$$\frac{dV_{c1}}{dt} = \left[\frac{1}{C_1}\left(i_L + \frac{1}{R}\left(-V_{c1} + v(t) - V_{c2}\right)\right)\right]$$

$$\frac{dx_1}{dt} = \frac{1}{C_1}x_2 - \frac{1}{RC_1}x_1 - \frac{1}{RC_1}x_3 + \frac{1}{RC_1}v(t)$$

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} -1/RC_1 & 1/C_1 & -1/RC_1 \\ 1/L & 0 & 0 \\ -1/RC_2 & 0 & -1/RC_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/RC_1 \\ 1/L \\ 1/RC_2 \end{bmatrix} v(t)$$

$$\overbrace{y(t)}^{V_o(t)} = V_{c2}$$

$$y = x_3$$

$$[y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\rightarrow$ (v) $i_{C_1} = i_L + \left(\frac{V_L - V_{C2}}{R}\right)$

$$i_{C_1} = i_L + \frac{1}{R}\Big[\underbrace{-V_{C1} + v(t)}_{V_L} - V_{C2}\Big]$$

$$\frac{dV_{C1}}{dt} = \left[\frac{1}{C_1}\left(i_L + \frac{1}{R}(-V_{C_1} + v(t) - V_{C2}\right)\right]$$

$$\frac{dx_1}{dt} = \frac{1}{C_1}x_2 - \frac{1}{RC_1}x_1 - \frac{1}{RC_1}x_3 + \frac{1}{RC_1}v(t)$$

$$\dot{X} = AX + BU$$
$$Y = CX + DU$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \end{bmatrix} = \begin{bmatrix} -1/RC_1 & 1/C_1 & -1/RC_1 \\ -1/L & 0 & 0 \\ -1/RC_2 & 0 & -1/RC_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1/RC_1 \\ 1/L \\ 1/RC_2 \end{bmatrix} v(t)$$

$$y(t) = \overset{V_0(t)}{V_{C2}}$$
$$y = x_3$$
$$[y] = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

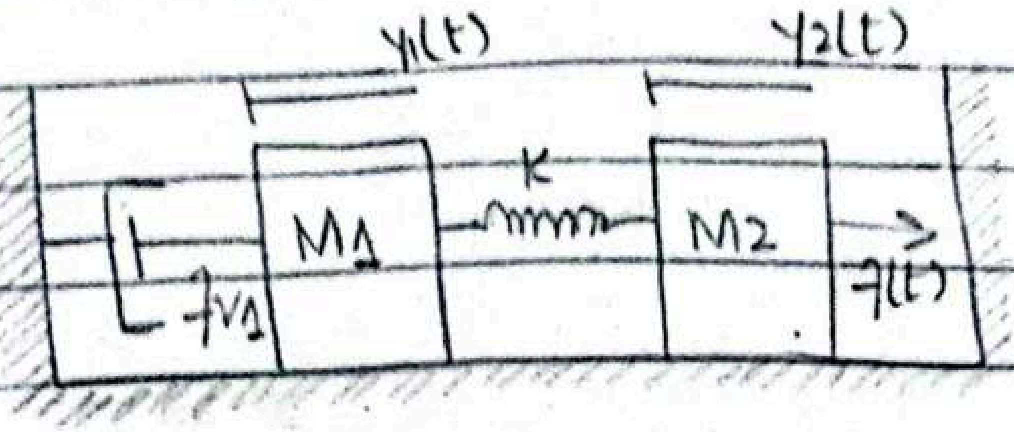
State Representation of Mechanical Systems

∴ is obtained from point of motion/displacement rather than energy storing elements.

∴ Take displacement rather and velocity at each point as state space variables.

→ if its a 2nd state equation

$$\begin{bmatrix}\text{num of}\\ \text{state variable}\\ \text{in equation}\end{bmatrix} = \begin{bmatrix}\text{equal to the}\\ \text{order of that}\\ \text{equation}\end{bmatrix}$$



→ $M_1 \frac{d^2y_1}{dt} + f_{v_1}\frac{dy}{dt} + k(y_1 - y_2) = 0$ —(i)

→ $M_2 \frac{d^2y_2}{dt^2} + k(y_2 - y_1) = f(t)$ —(ii)

$$y_1 = x_1 , \frac{dy_1}{dt} = x_2 \Rightarrow \frac{dx_1}{dt} = x_2 \Rightarrow \frac{dx^2}{dt} = \frac{d^2y_1}{dt^2}$$

$$y_2 = x_3 , \frac{dy_2}{dt} = x_4 \Rightarrow \frac{dx_3}{dt} \Rightarrow x_4 \Rightarrow \frac{dx_4}{dt} = \frac{d^2y_2}{dt^2}$$

$$\frac{d^2y_1}{dt^2} = \frac{-k}{M_1}y_1 + \frac{k}{M_1}y_2 - \frac{f_{v_1}}{M_1}\frac{dy_1}{dt}$$

$$\frac{dx_2}{dt} = \frac{-k}{M_1}\cdot x_1 + \frac{k}{M_2}x_3 - \frac{f_{v_1}}{M_1}x_2 \quad —*$$

Similarily

$$\frac{d^2y_2}{dt^2} = \frac{k}{M_2}y_1 - \frac{k}{M_2}y_2 + \frac{1}{M_2}f(t)$$

$$\frac{dx_4}{dt} = \frac{k}{M_2}x_1 - \frac{k}{M_2}x_3 + \frac{1}{M_2}f(t)$$

$$\begin{bmatrix} dx_1/dt \\ dx_2/dt \\ dx_3/dt \\ dx_4/dt \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -k/M_1 & -f_{v1}/M_1 & k/M_1 & 0 \\ 0 & 0 & 0 & 1 \\ k/M_2 & 0 & -k/M_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f(t)$$

$$y_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \qquad \therefore y_1 = x_1$$

$$y_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \qquad \therefore y_2 = x_3$$

State space representation from Transfer Fn

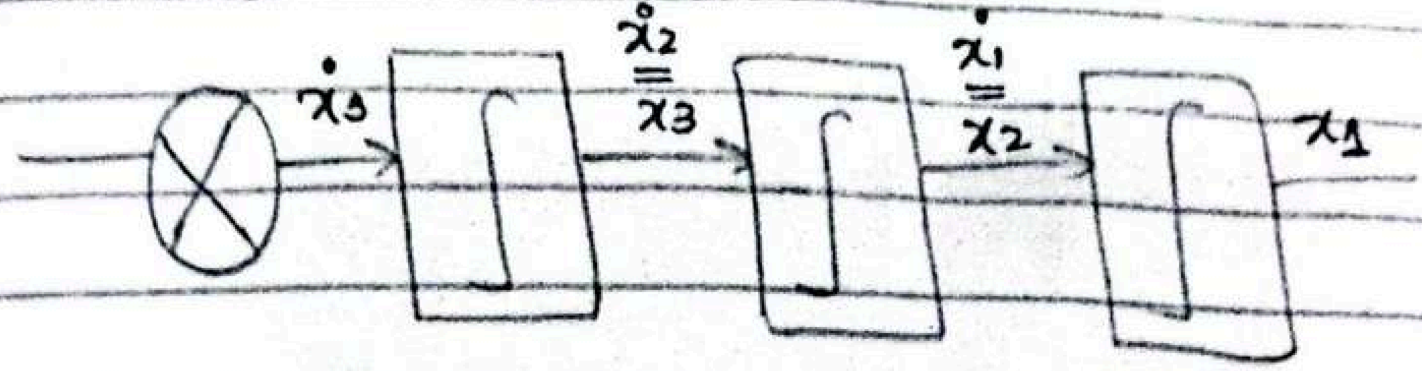
- The no. of state variables = the order of system
- Select set of state variables in such a way that each variable is derivative of prev variable

$$\frac{Cs}{Rs} = \frac{24}{s^3+9s^2+26s+24}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

1. No. of integrators will be equal to no. of state variables

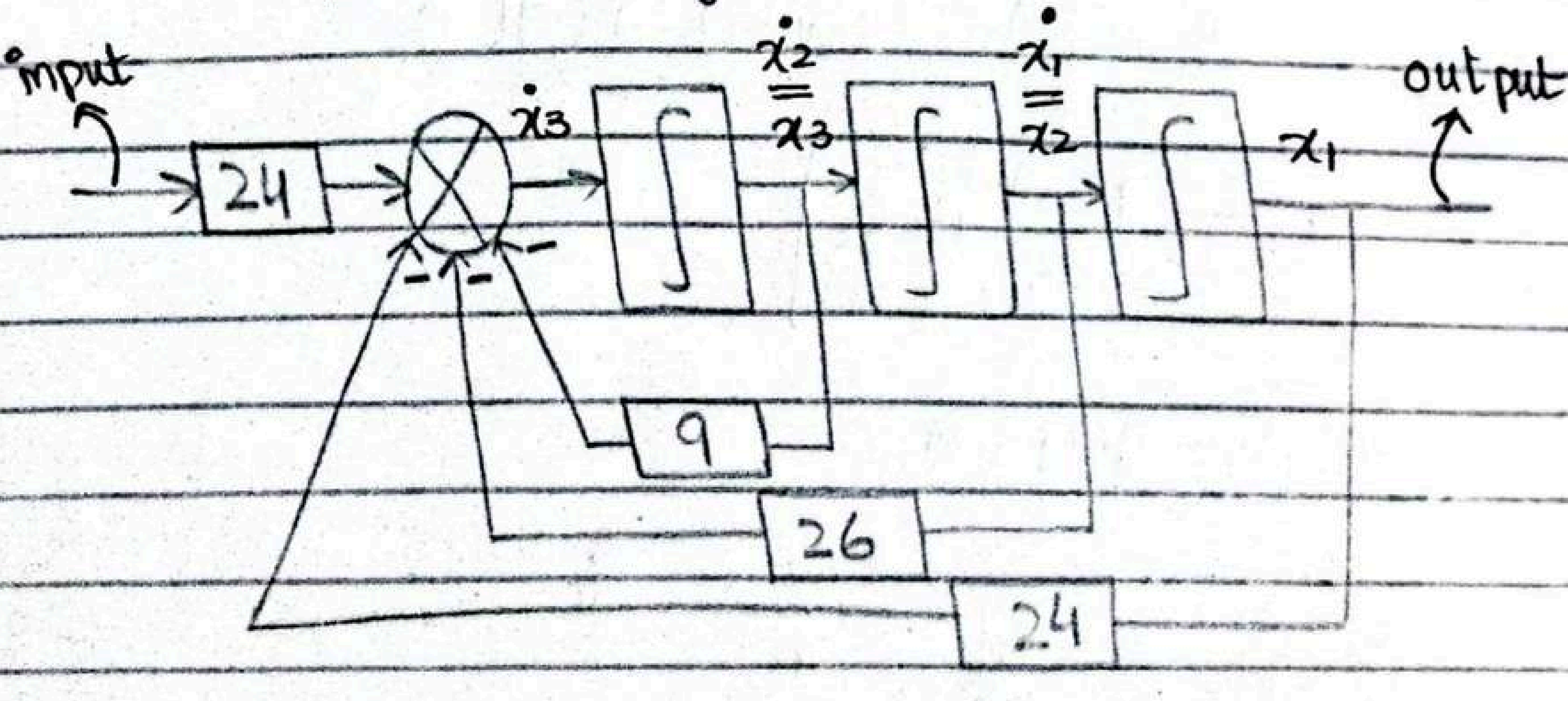


Now by matrix:

$$\frac{d^3c}{dt^3} + 9\frac{d^2c}{dt^2} + 26\frac{dc}{dt} + 24 = 24r$$

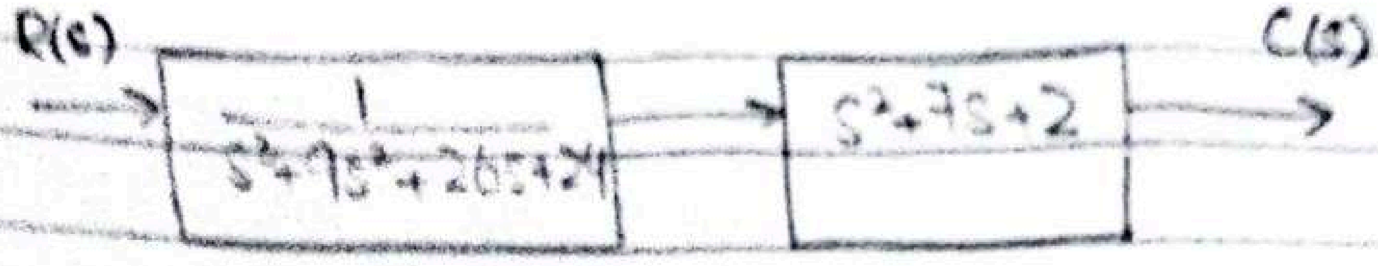
$$c = x_1, \frac{dc}{dt} = x_2, \frac{dx_2}{dt} = \frac{d^2c}{dt} = x_3$$

Now updated diagram will be



* transfer function is defined for having one-input and one output

$$\frac{C(s)}{R(s)} = \frac{s^2+7s+2}{s^3+9s^2+26s+24}$$



$$Y(s) = \frac{1}{s^3+9s^2+26s+24} \cdot (R(s))$$

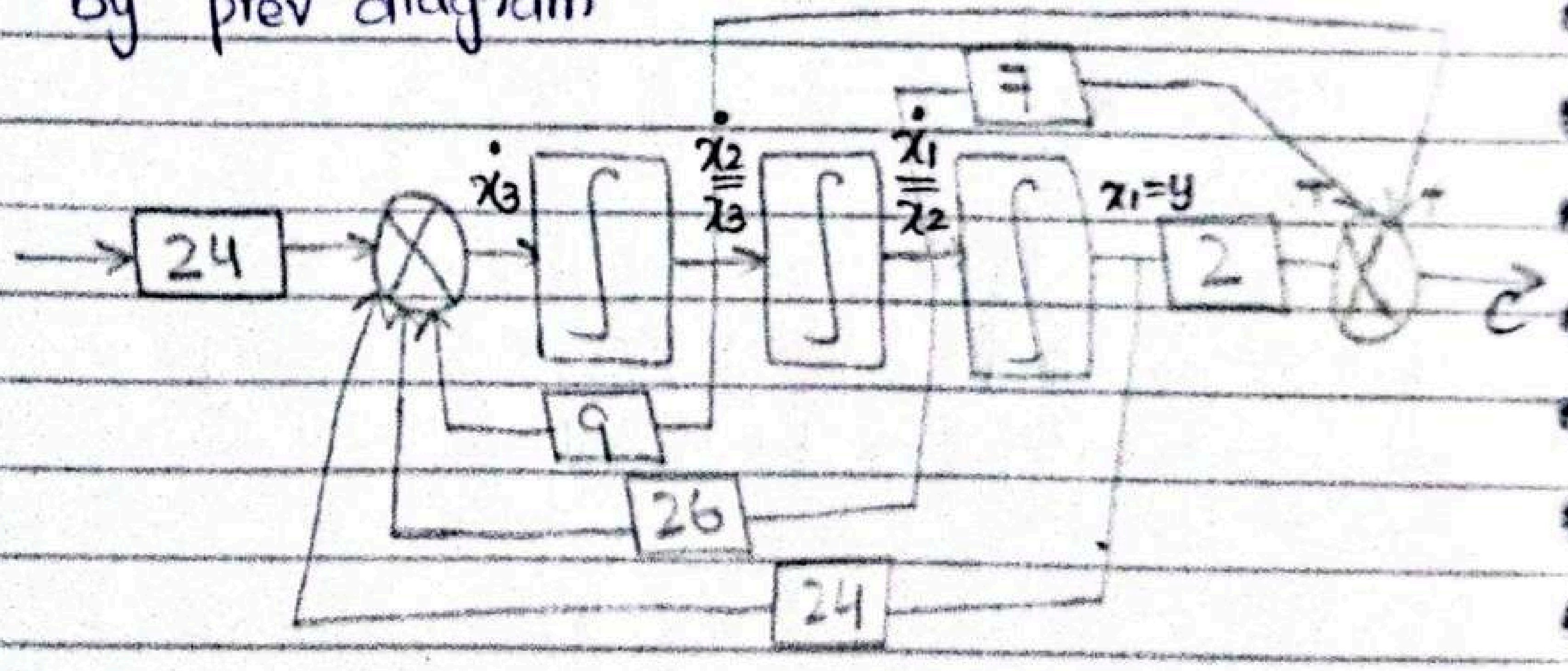
$y = x_1$; $\frac{dy_1}{dx} = x_2$; $\frac{d^2y}{dx^2} = x_3$

$$c(t) = \frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 2y(t)$$

↳ from $C(s) = (s^2+7s+2)(Y(s))$

$$C = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

by prev diagram



Converting State space Model into T.F :

$$\dot{X} = AX + BU$$

$$sX(s) = AX(s) + BU(s)$$

$$Y = CX + DU$$

$$Y(s) = CX(s) + DU(s)$$

$$sX(s) - AX(s) = BU(s)$$

$$X(s)[sI - A] = BU(s)$$

$$X(s) = [sI - A]^{-1} BU(s)$$

$$Y(s) = C(sI - A)^{-1} BU(s) - DU(s)$$

$$= [C(sI - A)^{-1} B + D] U(s)$$

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1} B + D$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow sI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s+3 \end{bmatrix}$$

$$(sI-A)^{-1} = \frac{Adj(sI-A)}{\det(sI-A)}$$

$$\det(sI-A) = s(s(s+3)+2)+1[0+1]$$
$$= s^3+3s^2+2s+1$$

$$Adj\,A = [\text{Matrix of co-factors}]^T$$

$$\text{Matrix of co-factors} = \begin{bmatrix} s(s+3)+2 & -1 & -s \\ s+3 & s^2+3s & -(2s+1) \\ 1 & s & s^2 \end{bmatrix}$$

$$Adj(sI-A) = \begin{bmatrix} s^2+3s+2 & s+3 & 1 \\ -1 & s^2+3s & s \\ -s & -(2s+1) & s^2 \end{bmatrix}$$

$$T.F = C(sI-A)^{-1}B+D$$
$$= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \frac{1}{\Delta} \begin{bmatrix} s^2+3s+2 & s+3 & 1 \\ -1 & s^2+3s & s \\ -s & -(2s+1) & s^2 \end{bmatrix} \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

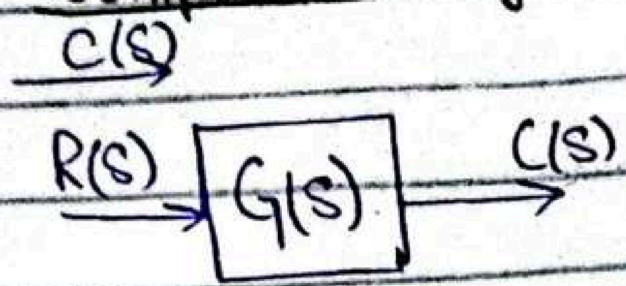
$$= \frac{1}{\Delta} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10(s^2+3s+2) \\ -10 \\ -10s \end{bmatrix}$$

$$= \frac{10(s^2+3s+2)}{s^3+3s^2+2s+1}$$

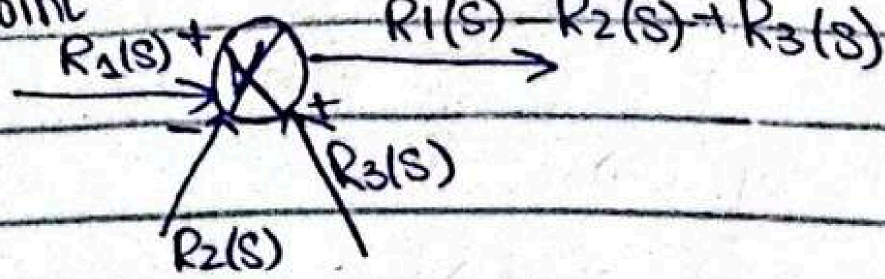
Chapter no 5

BLOCK DIAGRAM REDUCTION

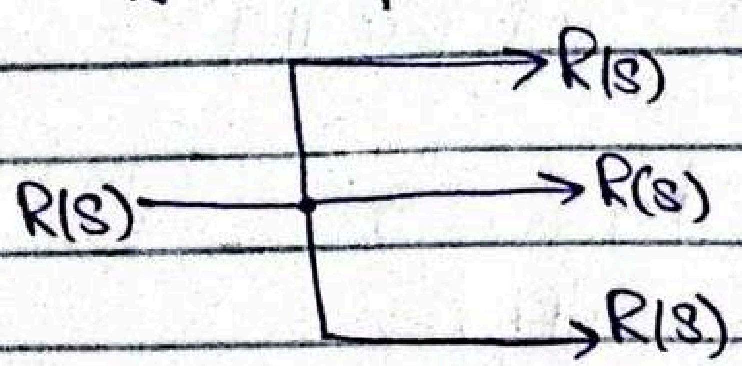
Components of a block diagram

- 

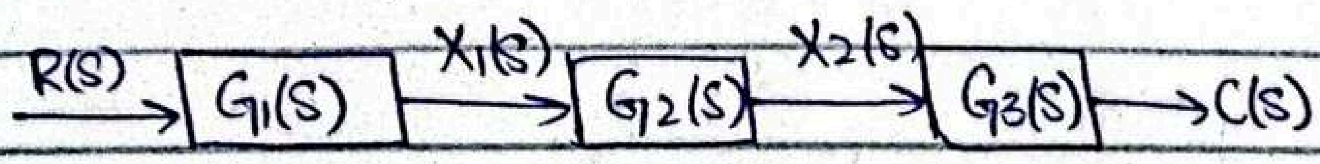
- Summing point



- Pick off point / Branch point



→ CASCADE System:



$$X_1(S) = G_1(S)R(S)$$

$$X_2(S) = G_2(S)\,X_1(S) = G_2(S)G_1(S)\,R(S)$$

$$C_S = G_3(S)\,X_2(S)$$

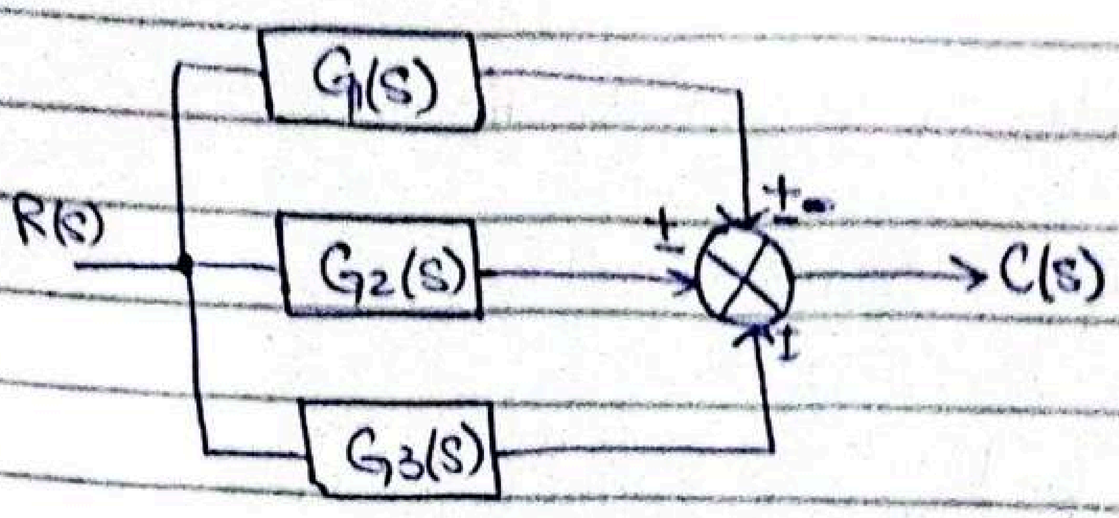
$$= G_3(S)\,G_2(S)G_1(S)\,R(S)$$

$$\frac{C(S)}{R(S)} = G_3(S)\,G_2(S)\,G_1(S)\,R(S) \quad \text{(Transfer function)}$$

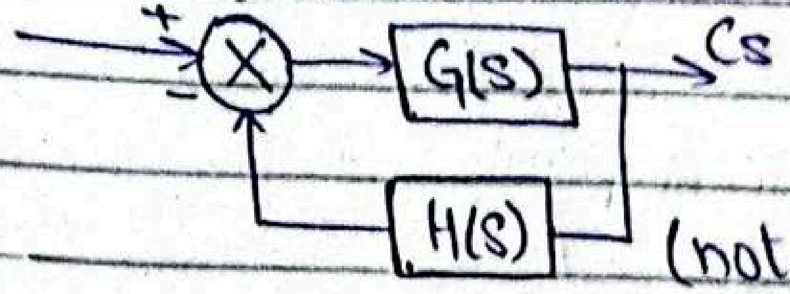
Parallel sub-systems

↓

i/p and o/p at the same point

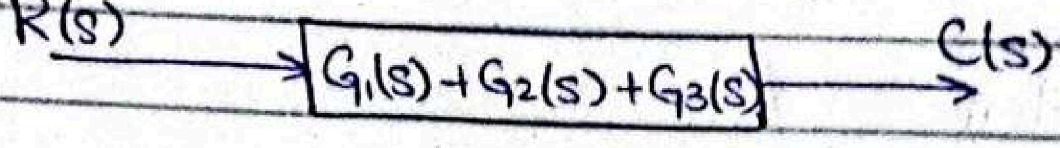


feedback system

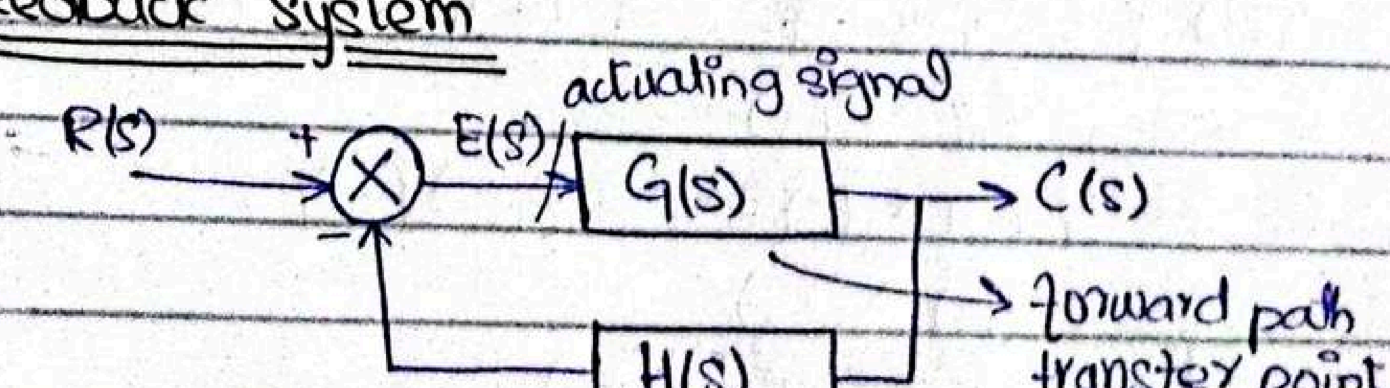


(not parallel)

$$C(s) = G_1(s)\,R(s) + G_2(s)\,R(s) + G_3(s)\,R(s)$$
$$= \left[G_1(s) + G_2(s) + G_3(s)\right] R(s)$$



Feedback System



∴ actuator – signal at summing point

$$C(s) = G(s)\,E(s) \quad (i)$$

$$E(s) = R(s) - B(s)$$

$$B(s) = H(s)G(s)$$

$$E(s) = R(s) - H(s)G(s)$$

$$\Rightarrow (i) \quad C(s) = G(s)\cdot\left[R(s) - H(s)G(s)\right]$$

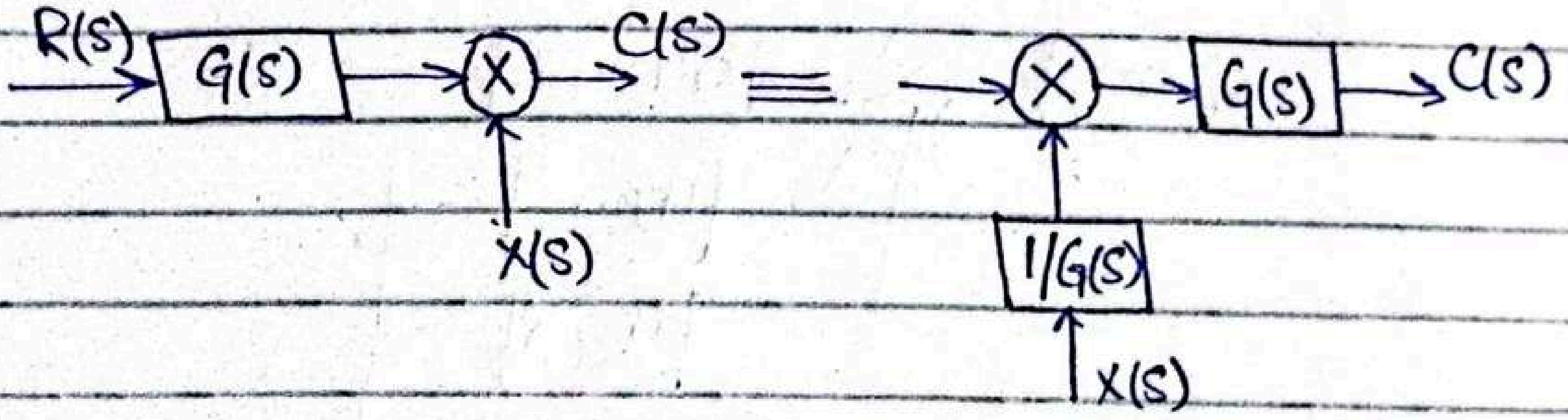
$$C(s) + G(s)H(s)\,C(s) = G(s)R(s)$$

$$C(s)\left[1 + G(s)H(s)\right] = G(s)R(s)$$

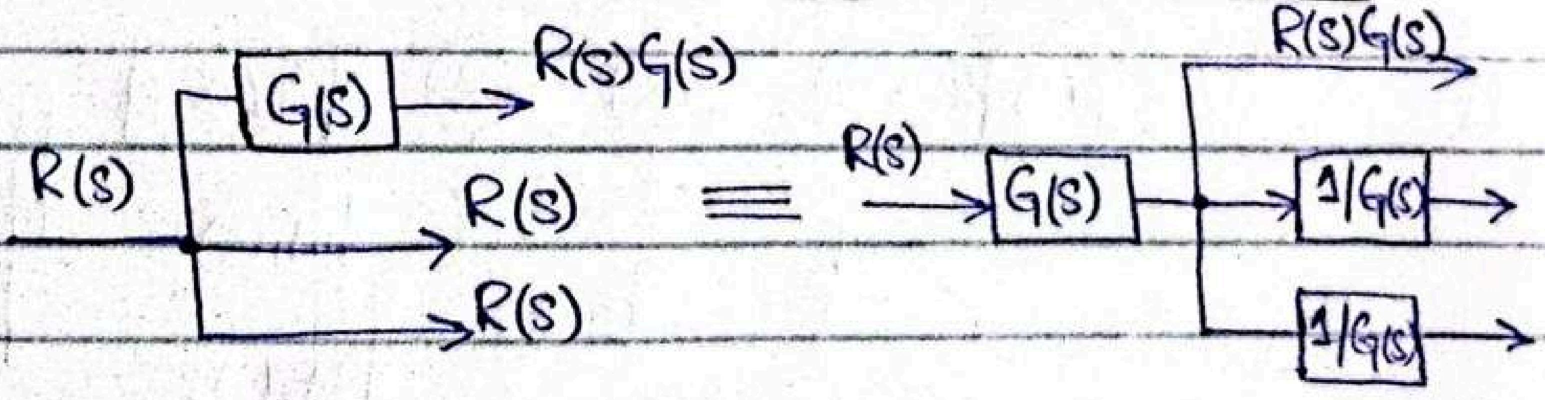
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \mp \underbrace{G(s)H(s)}_{\text{open loop transfer function}}}$$

(closed Loop transfer function)

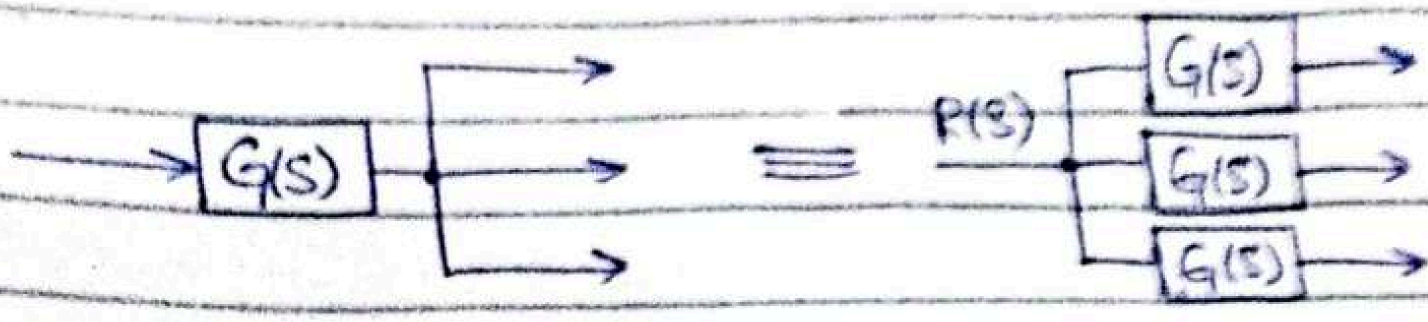
→ **Moving a transfer function to the Left of summing point**



→ **Moving a T.F to the Left of pick off point**

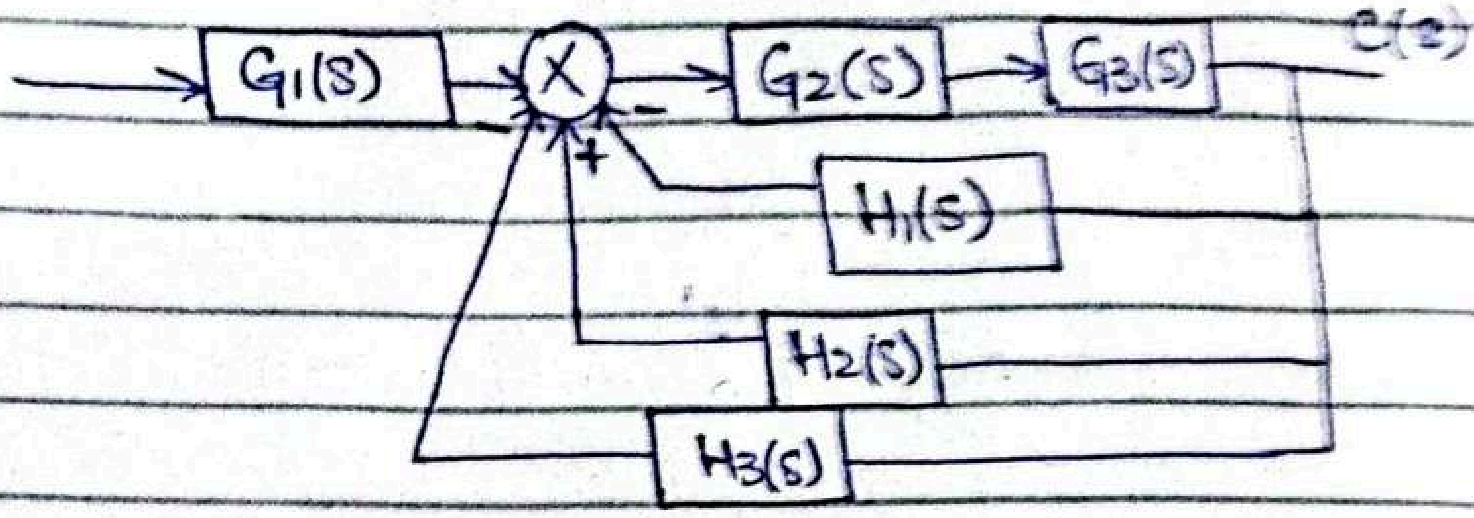


→ Moving a TF to the right of pick off point:

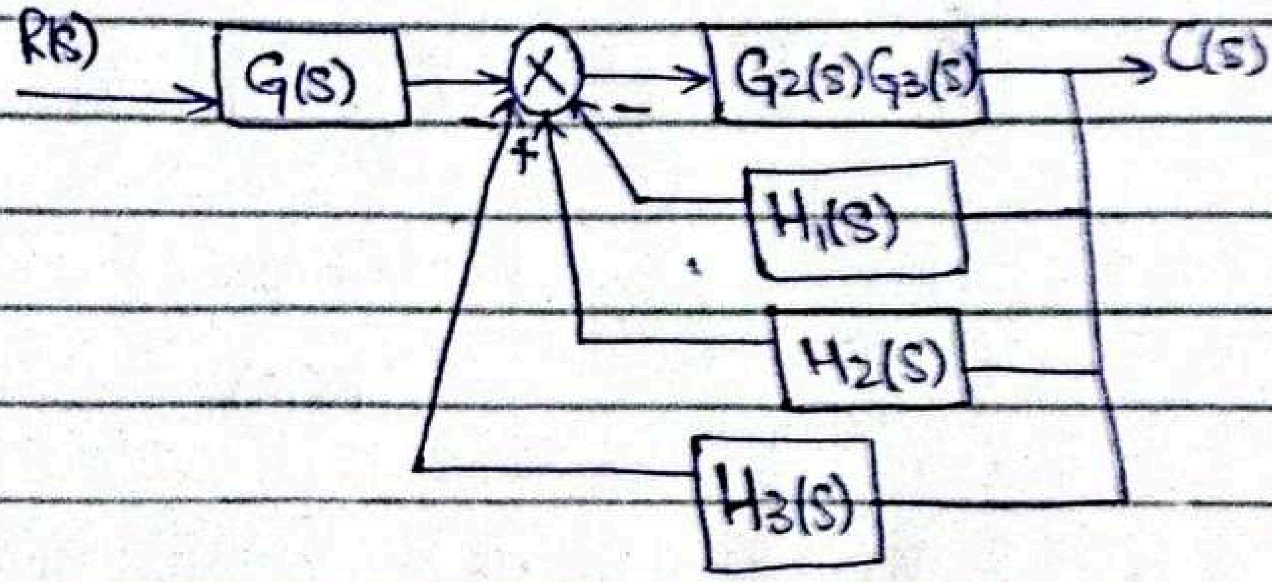


Example #

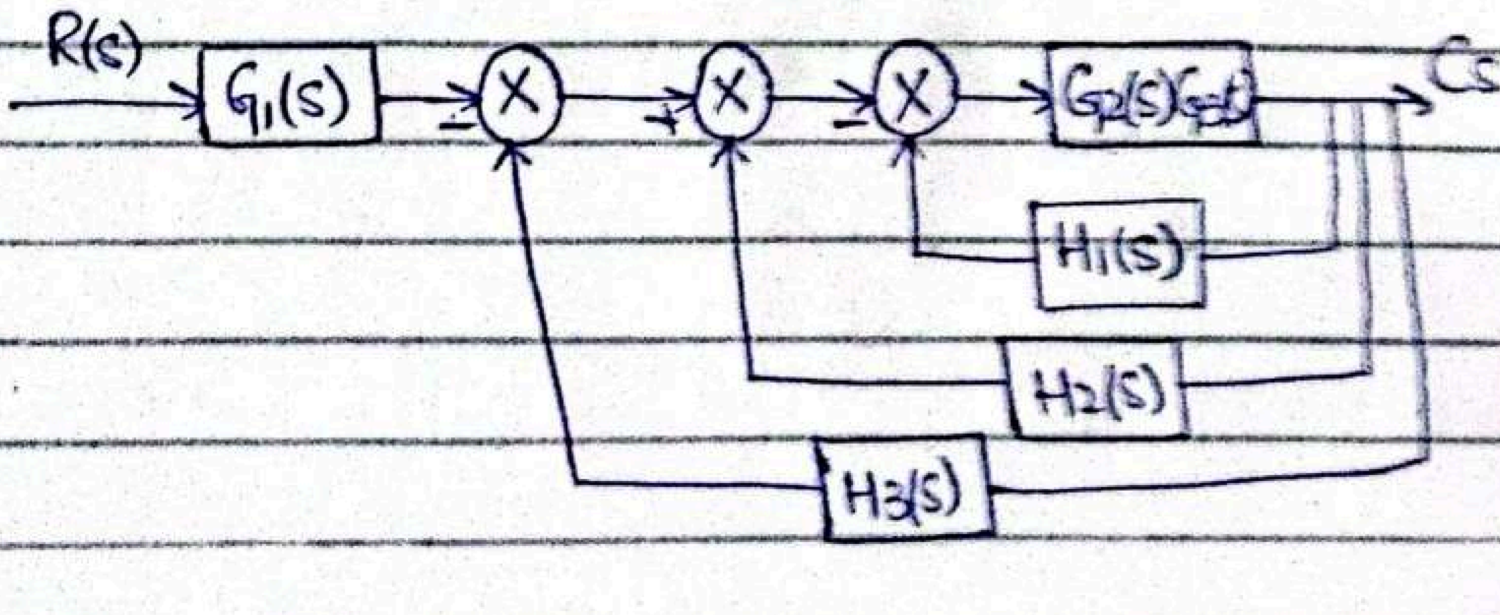
Method I



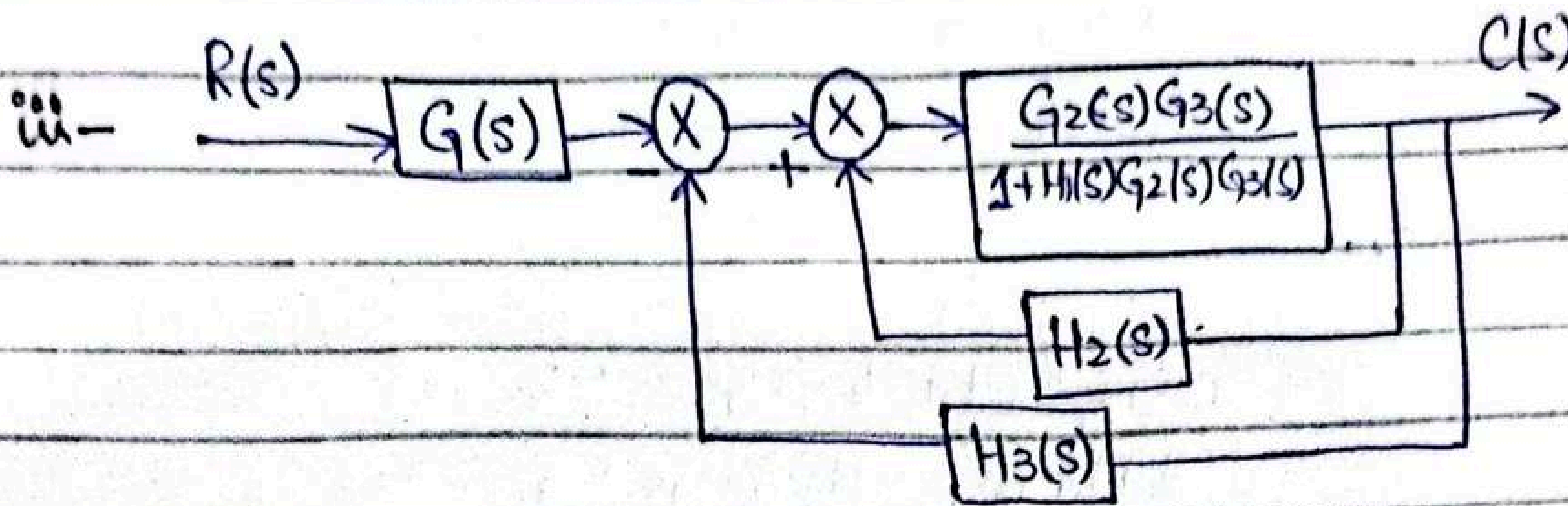
i-



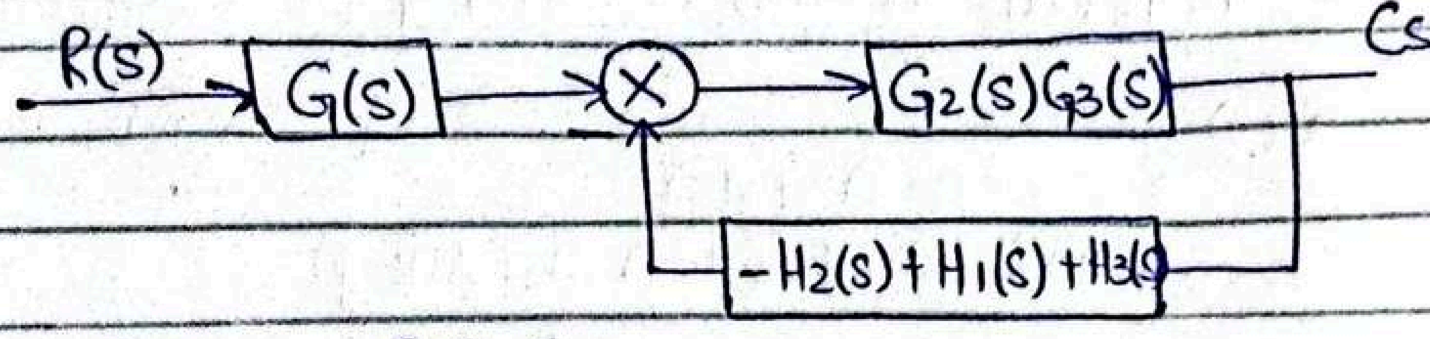
ii-



iii-



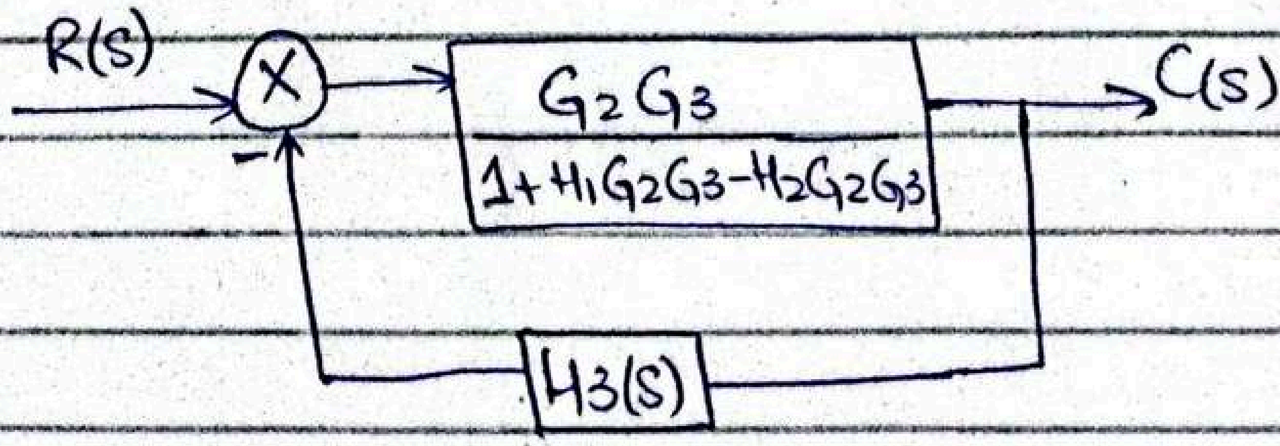
Method II



$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{1+G_2G_3(H_1-H_2+H_3)}$$

iv- $$\frac{G_2G_3/(1+H_1G_2G_3)}{1-\left(\frac{G_2G_3}{1+H_1G_2G_3}\right)\times H_2} = \frac{G_2G_3/\cancel{(1+H_1G_2G_3)}}{\frac{1+H_1G_2G_3-G_2G_3H_2}{\cancel{1+H_1G_2G_3}}}$$

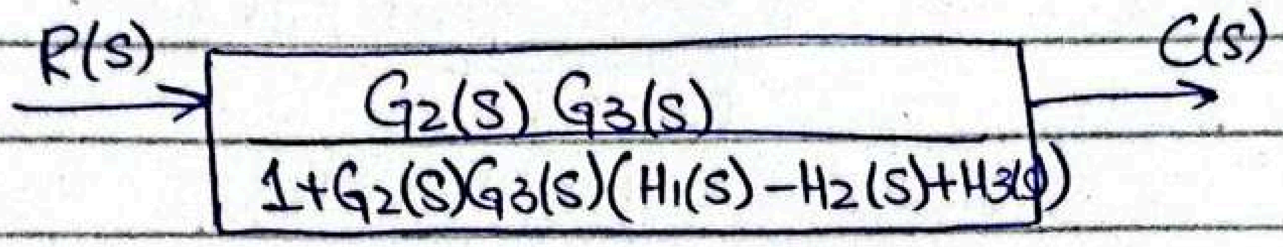
$$= \frac{G_2G_3}{1+H_1G_2G_3-H_2G_2G_3}$$



$$\Rightarrow \frac{G_2G_3/(1+H_1G_2G_3-H_2G_2G_3)}{1+\left(\frac{G_2G_3}{1+H_1G_2G_3-H_2G_2G_3}\right)\times H_3}$$

$$= \frac{\dfrac{G_2G_3}{\cancel{(1+H_1G_2G_3-H_2G_2G_3)}}}{\dfrac{1+H_1G_2G_3-H_2G_2G_3+G_2G_3H_3}{\cancel{1+H_1G_2G_3-H_2G_2G_3}}}$$

$$= \frac{G_2G_3}{1+G_2G_3(H_1-H_2+H_3)} = \frac{G_2G_3}{1+G_2G_3(H_1-H_2+H_3)}$$

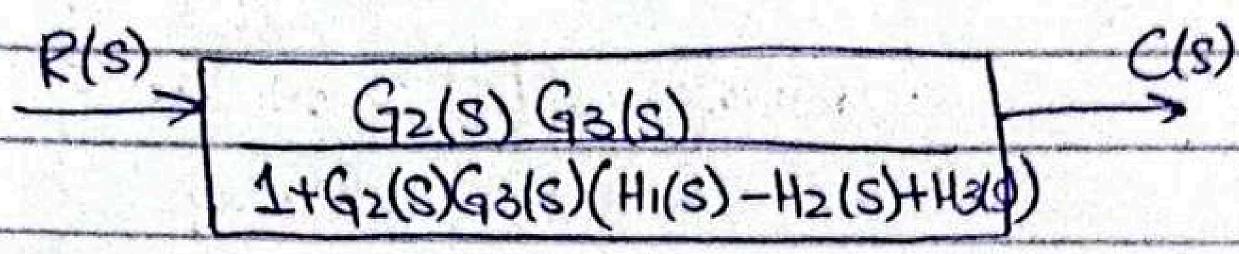


$$\frac{C(S)}{R(S)} = \frac{G_2(S)G_3(S)}{1+G_2(S)G_3(S)(H_1(S)-H_2(S)+H_3(S))}$$

$$V - \frac{G_2G_3/(1+H_1G_2G_3-H_2G_2G_3)}{1+\left(\dfrac{G_2G_3}{1+H_1G_2G_3-H_2G_2G_3}\right)\times H_3}$$

$$= \frac{G_2G_3/\cancel{(1+H_1G_2G_3-H_2G_2G_3)}}{\dfrac{1+H_1G_2G_3-H_2G_2G_3+G_2G_3H_3}{\cancel{1+H_1G_2G_3-H_2G_2G_3}}}$$

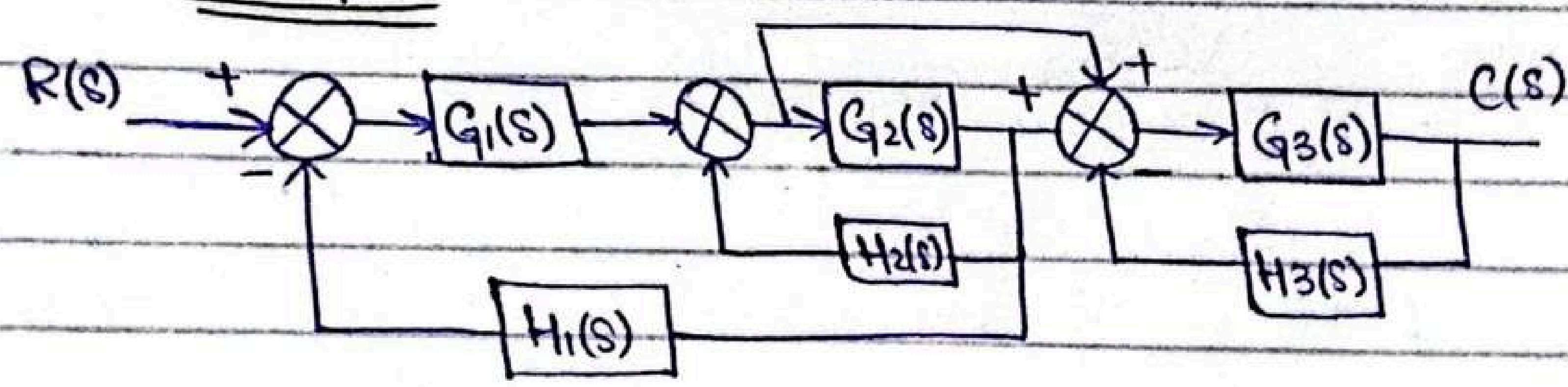
$$= \frac{G_2G_3}{1+G_2G_3(H_1-H_2+H_3)} = \frac{G_2G_3}{1+G_2G_3(H_1-H_2+H_3)}$$



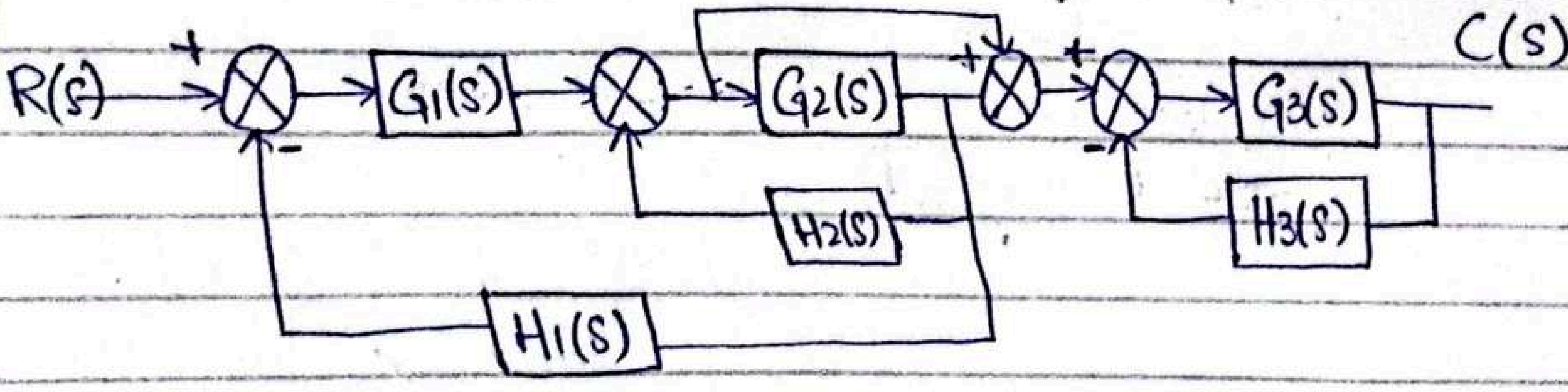
$$\frac{C(S)}{R(S)} = \frac{G_2(S)G_3(S)}{1+G_2(S)G_3(S)(H_1(S)-H_2(S)+H_3(S))}$$

(12)

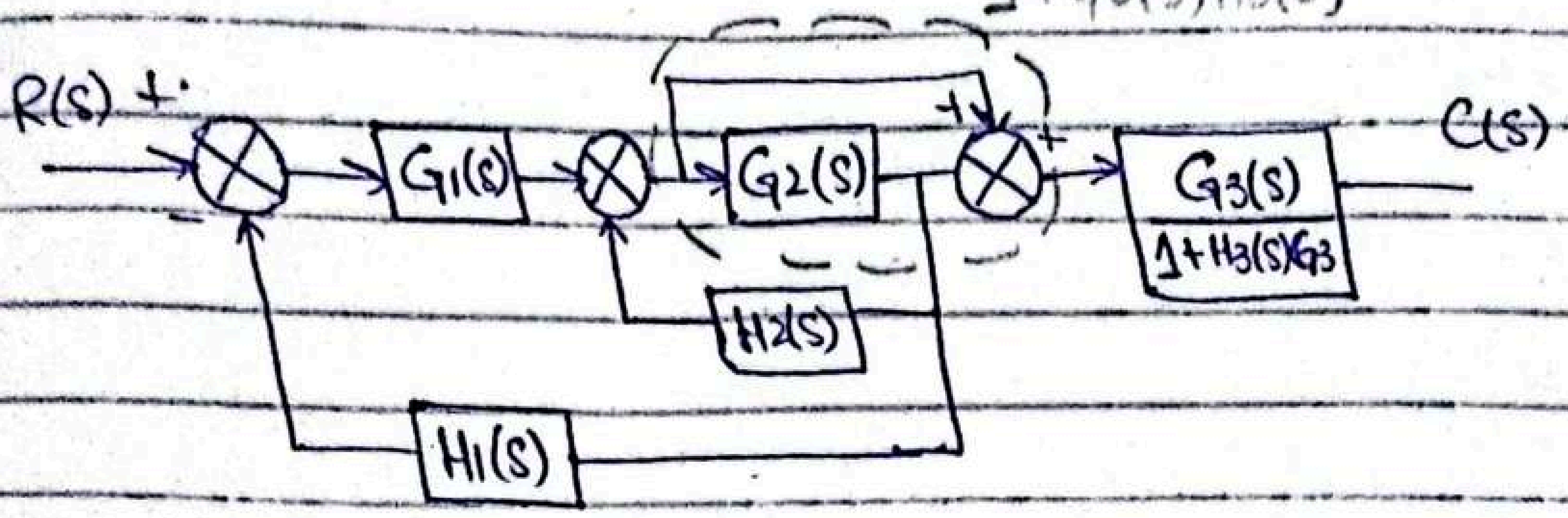
Example:



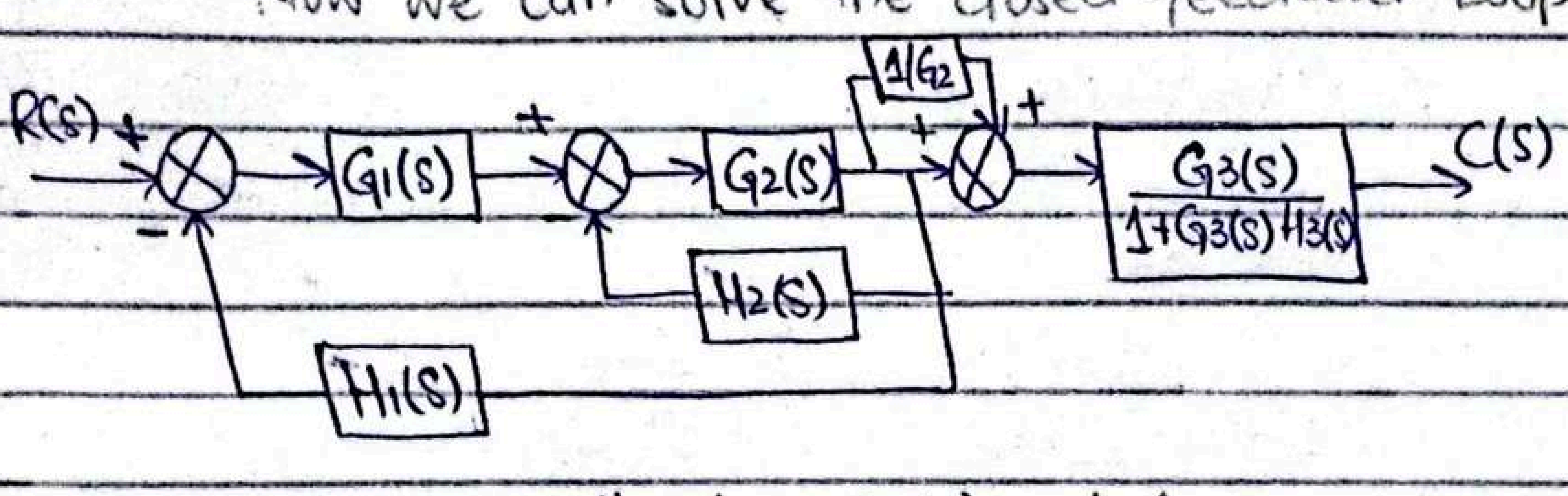
∴ we will put an adder before $G_3(s)$



Now, $G_3(s)$ and $H_3(s)$ are in -ve Loop
we can solve them
So, it will be $= \dfrac{G_3(s)}{1+G_3(s)H_3(s)}$

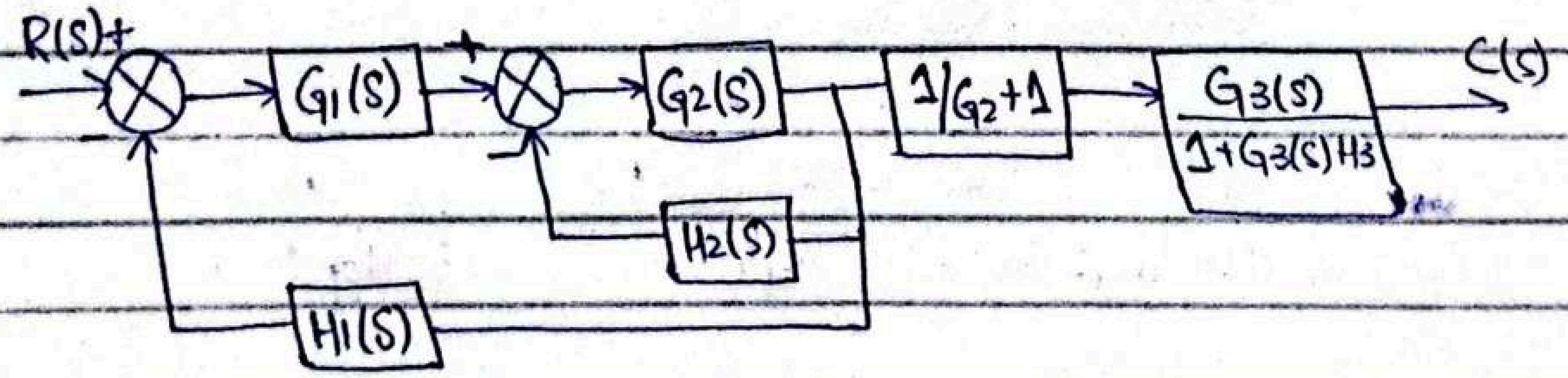


Now we can solve the closed feedback Loop



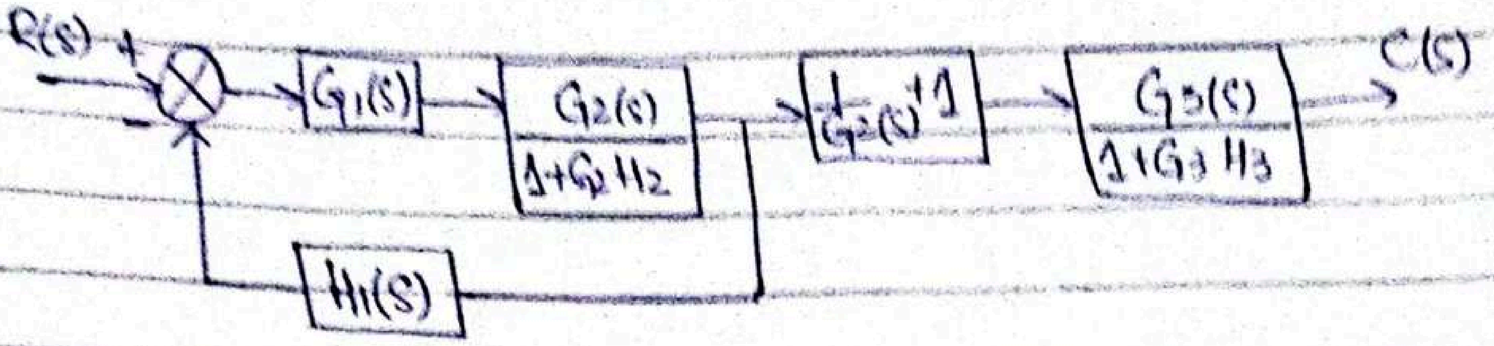
Now we will solve $G_2(s)$ and $\dfrac{1}{G_2(s)}$

∴ We cannot interchange summing and pick off point



Now, G_2 and H_2 are in -ve feedback Loop So,

$$= \frac{G_2(s)}{1+G_2(s)H_2(s)}$$



Now $G_1(s)$ and $\frac{G_2(s)}{1+G_2H_2}$ are in series
then they're in negative feedback with $H_1(s)$
and then they're in series with next two blocks.

Solving the expression:

$$= G_1(S)\left(\frac{G_2(S)}{1+G_2H_2}\right) \quad \therefore \text{in series}$$

$$= \frac{G_1G_2}{1+G_2H_2} \quad \text{(negative feedback with } H_1(S)\text{)}$$

$$= \frac{G_1G_2/1+G_2H_2}{1+\frac{G_1G_2}{1+G_2H_2}\cdot H_1} = \frac{G_1G_2/1+G_2H_2}{\frac{1+G_2H_2+G_1G_2H_2}{1+G_2H_2}}$$

$$= \frac{G_1G_2}{1+G_2H_2+G_1G_2H_2}$$

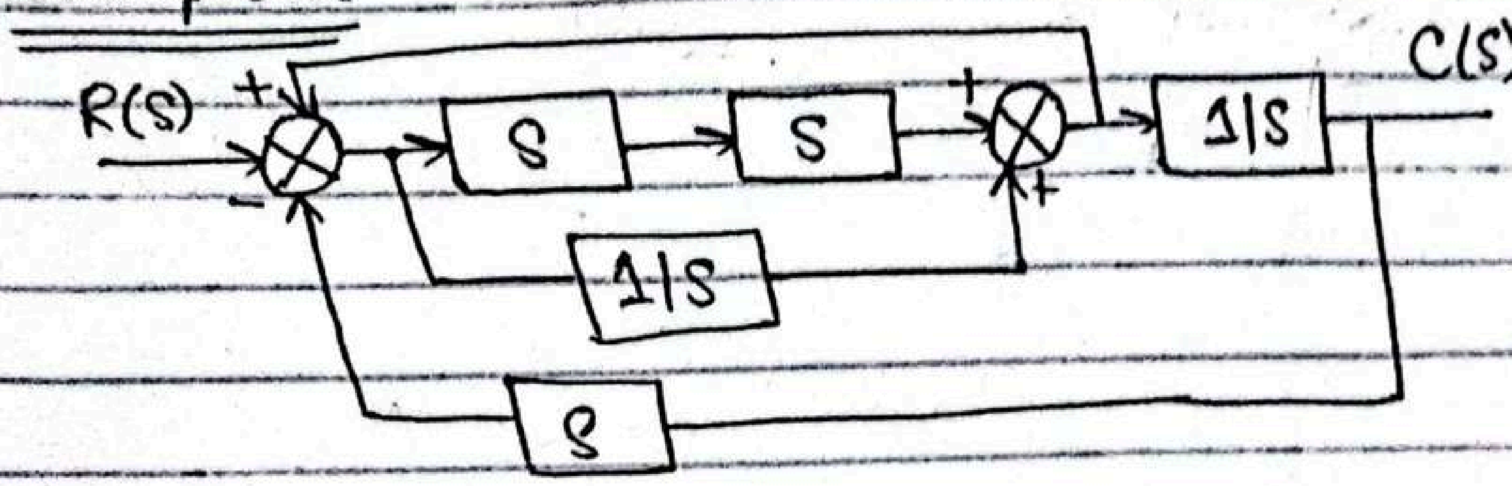
Now solving all three blocks in series

$$= \frac{G_1G_2}{1+G_2H_2+G_1G_2H_2} \cdot \frac{1+G_2}{G_2} \cdot \frac{G_3}{1+G_3H_3}$$

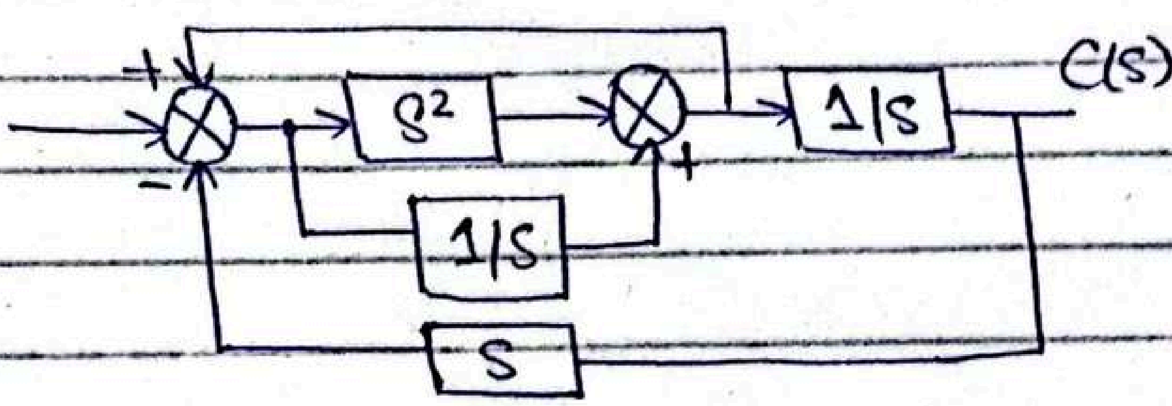
$$= \frac{G_1G_3(1+G_2)}{(1+G_2H_2+G_1G_2H_2)(1+G_3H_3)} = \frac{C(S)}{R(S)}$$

Series $\longrightarrow$ Multiply
Parallel $\longrightarrow$ Addition
Negative feedback $\longrightarrow$ $\dfrac{G_1(s)}{1+G_1(s)(H_1(s))}$
Unity close Loop $\longrightarrow$ $\dfrac{G_1(s)}{1+G_1(s)}$

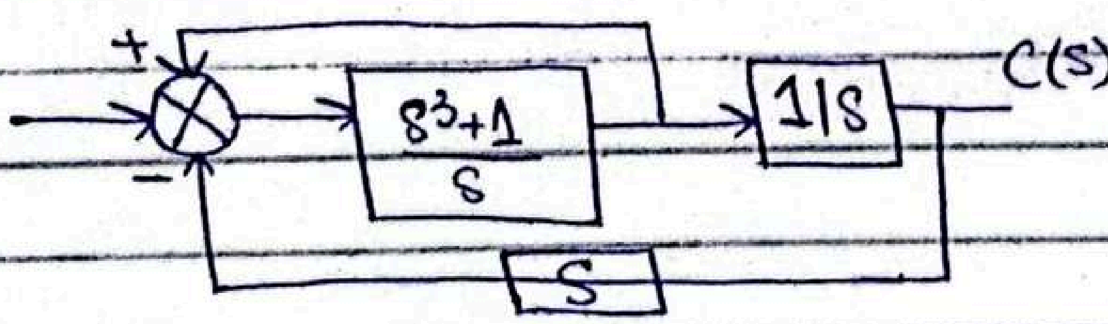
Example 2:



$= (S)(S) = S^2$ (series)

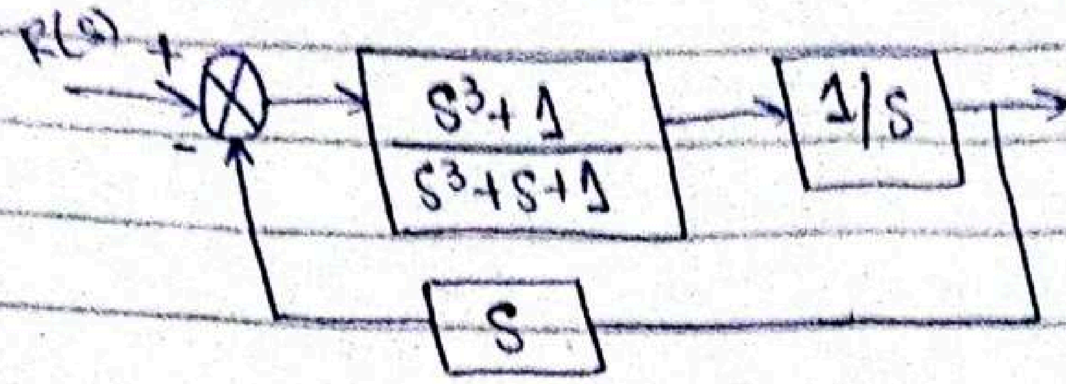


$= S^2 + \dfrac{1}{S} = \dfrac{S^3+1}{S}$ (parallel)

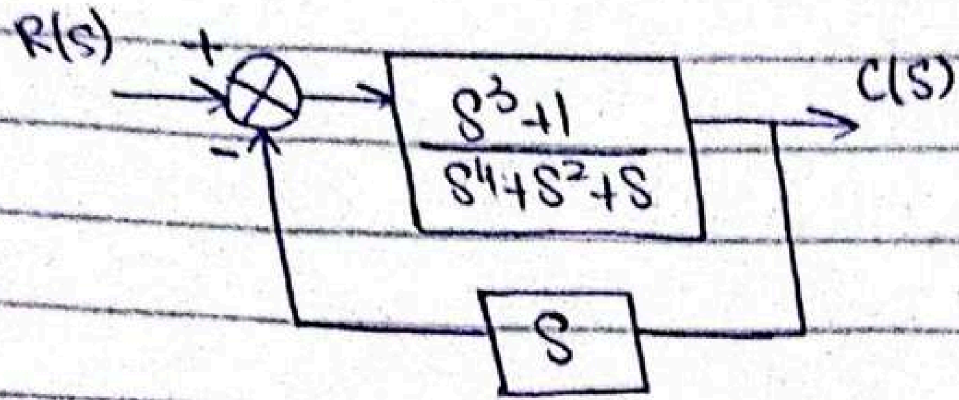


Solving closed Loop

$$= \frac{S^3+1/S}{1+S^3+1/S} = \frac{S^3+1/\cancel{S}}{1+S^3+S/\cancel{S}} = \frac{S^3+1}{S^3+S+1}$$



$$=\left(\frac{s^3+1}{s^3+s+1}\right)\left(\frac{1}{s}\right)=\frac{s^3+1}{s^4+s^2+s}\quad \text{(series)}$$



$$=\frac{s^3+1/s^4+s^2+s}{1+\left(\frac{s^3+1}{s^4+s^2+s}\right)(s)}=\frac{s^3+1/s^4+s^2+s}{\frac{s^4+s^2+s+(s^3+1)s}{s^4+s^2+s}}$$

$$=\frac{s^3+1}{2s^4+s^3+2s}$$

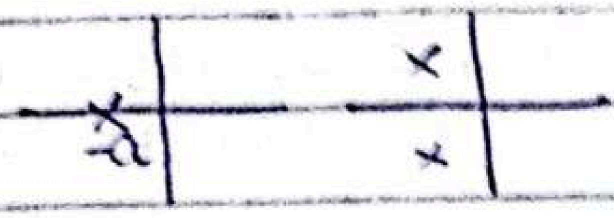
So,

$$\frac{R(s)}{C(s)}=\frac{s^3+1}{2s^4+s^3+2s}$$

Chapter no 04

Time Analysis of first order system

$$\frac{C(s)}{R(s)} = G(s) = \frac{k}{s+a}$$



Total Response = Forced Response + Natural Response

Step response:

$$C(s) = \frac{k}{s(s+a)}$$

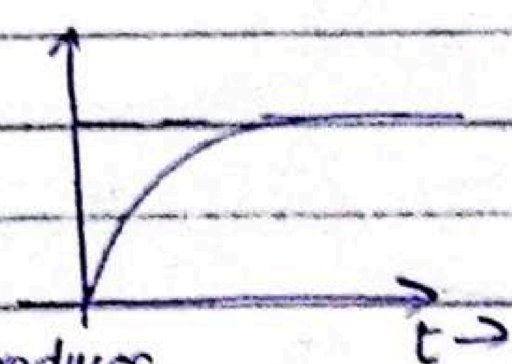
$$= \underbrace{\frac{k/a}{s}}_{\text{forced}} - \underbrace{\frac{k/a}{s+a}}_{\text{natural}}$$

∴ the output due to poles is natural

∴ the output due to poles of input is forced

∴ In LTI, complex poles always exist in conjugate pairs.

$$c(t) = \frac{k}{a}\left(1 - e^{-at}\right)$$



Every real pole in s plane introduces an exp term in final response. the rate of decay or fall depends on the location pole.

$$G(s) = \frac{s+2}{s+5} \;;\; C(s) = \frac{s+2}{s(s+5)}$$

$$= \frac{A}{s} + \frac{B}{s+5}$$

$$= \frac{2/5}{s} - \frac{3/5}{s+5}$$

$$c(t) = \frac{2}{5}u(t) - \frac{3}{5}e^{-5t}u(t)$$

Transient Response of first order system:

1- **Time Constant:-**

Time for step response to reach 63% of its final value.

OR

Time for e^{-at} to decay to 37% of its initial value.

2- **Rise Time:-**

Time for the waveform to go from 0.1 – 0.9 (10 – 90%) of its final value

3- **Settling Time:-**

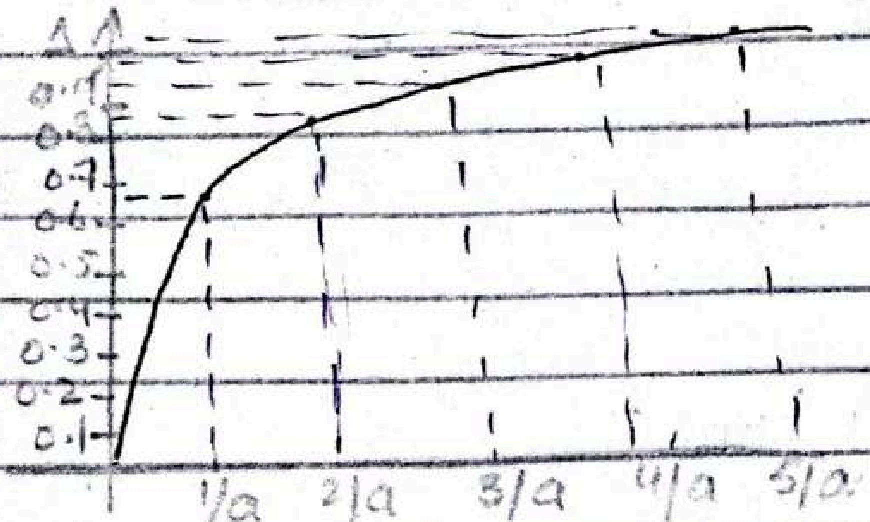
Time for the response to reach and stay within 2% of its final value.

$G(s) = \frac{k}{s+a}$; $c(t) = \frac{k}{a}(1-e^{-at})$

Assume $k=a$

$G(s) = \frac{a}{s+a}$; $C(s) = \frac{a}{s(s+a)}$

$$\boxed{c(t) = (1-e^{-at})u(t)}$$



→ $c(t)$ at $t = 1/a$

$c(t) = 1 - e^{-a(1/a)}$

$= 0.623$

→ $t = 2/a$

$= 1 - e^{-a(2/a)} = 0.864$

→ $t = 3/a$

$= 0.950$

→ $t = 4/a$

$= 0.981$

→ $t = 5/a$

$= 0.993$

1st order system = Time constant

$= 1/a$

∴ a is the location of pole

↳ For Rise time:

$T_r = t_2 - t_1$

↳ $c(t) = 1 - e^{-at}$

$0.1 = 1 - e^{-at}$

$e^{-at} = 1 - 0.1$

$-at = \ln(1-0.1)$

$-at = -0.105$

$$\boxed{t_1 = \frac{0.105}{a}}$$

$\hookrightarrow c(t) = 1-e^{-at}$

$$0.9 = 1-e^{-at}$$

$$e^{-at} = 1-0.9$$

$$\ln(e^{-at}) = \ln(1-0.9)$$

$$-at = -2.302$$

$$\boxed{t_2 = \frac{2.302}{a}}$$

$$T_r = t_2 - t_1 = \frac{2.302}{a} - \frac{0.105}{a} = \frac{2.2}{a}$$

→ <u>For settling time:</u>

$$c(t) = 1-e^{-at}$$

$$0.98 = 1-e^{-at}$$

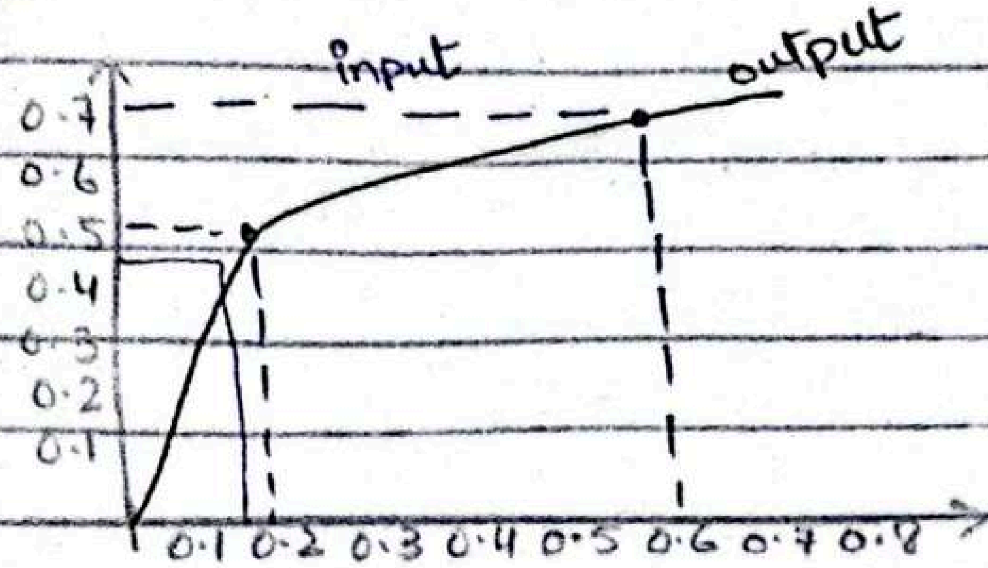
$$e^{-at} = 1-0.98$$

$$-at = -3.91$$

$$\boxed{t = \frac{3.91}{a} \approx \frac{4}{a}}$$

∴ It will take 4 time constants to reach the steady state.

Example:



Final value $= 0.72$

$0.63 \times 0.72 = 0.45$

Time constant $= \dfrac{1}{a} = 0.15 \Rightarrow a = 6.66$

$$c(t) = \frac{k}{a}\left(1 - e^{-at}\right)$$

$$\frac{K}{a} = \text{final value}$$

$$K = 0.72 \times a = 0.72 \times 6.6 = 4.75$$

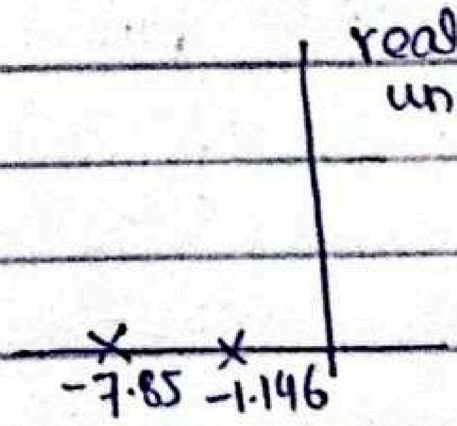
$$G(s) = \frac{4.75}{s + 6.6}$$

Transient Response of second order system

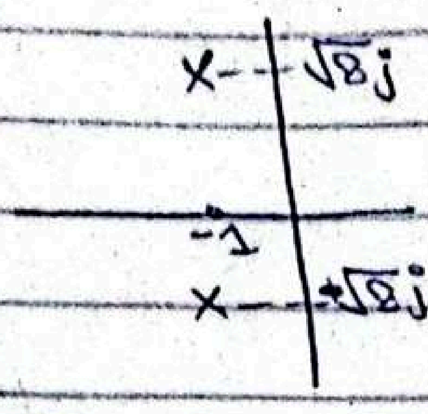
$$G_1(s) = \frac{9}{s^2 + 9s + 9}$$

$$= \frac{9}{(s + 7.854)(s + 1.146)}$$

real & unequal

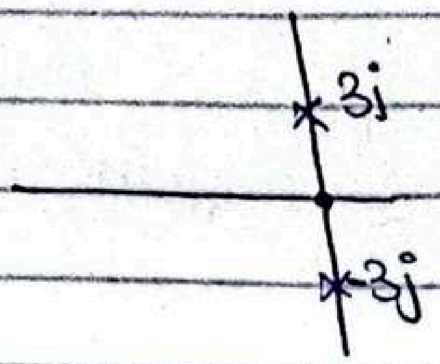


→ $G_2(s) = \dfrac{9}{s^2+2s+9} = \dfrac{9}{(s+1+\sqrt{8}j)(s+1-\sqrt{8}j)}$



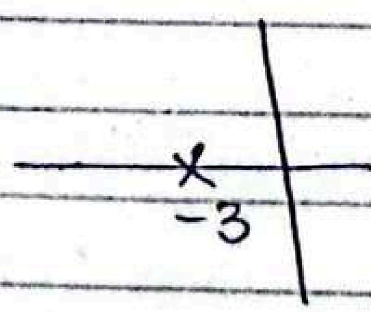
Complex Conjugate pair

→ $G_3(s) = \dfrac{9}{s^2+9} = \dfrac{9}{(s+3j)(s-3j)}$



poles on imaginary axis

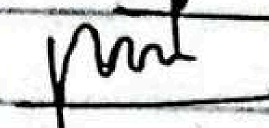
→ $G_4(s) = \dfrac{9}{s^2+6s+9} = \dfrac{9}{(s+3)^2}$

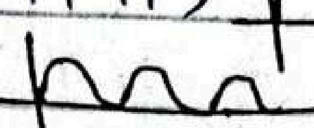


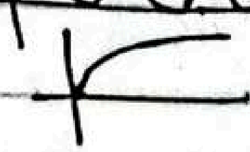
real & unequal

⟹ By partial fraction expansion

$c_1(t) = 1 + 0.171e^{-7.854t} - 1.171e^{-1.146t}$ 

$c_2(t) = 1 - 1.06e^{-t}\cos(\sqrt{8}t - 19.47°)$ 

$c_3(t) = 1 - \cos 3t$ 

$c_4(t) = 1 - 3te^{-3t} - e^{-3t}$ 

ω_n = natural frequency
ζ = damping rate
∴ Damping is actually the rate of decay and fall.

ζ = damping ratio

∴ Damping is actually the rate of decay and fall.

Second Order System

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n = Natural frequency

ζ = Damping ratio

$\zeta = 0 \Rightarrow$ undamped system

$0 < \zeta < 1 \Rightarrow$ underdamped system

$\zeta = 1 \Rightarrow$ critically damped

$\zeta > 1 \Rightarrow$ overdamped system

$$\rightarrow \frac{C(s)}{R(s)} = G(s) = \frac{9}{s^2 + 9s + 9} \qquad R(s) = \frac{1}{s}$$

$r(t) = u(t)$

$$C(s) = \frac{9}{s(s+7.854)(s+1.146)}$$

$$\hookrightarrow c(t) = 1 + 0.17e^{-7.854t} - 1.171e^{-1.146t}$$

∴ consider ω_n^2 from denominator

$\omega_n^2 = 9 \Rightarrow \omega_n = 3$

$2\zeta\omega_n = 9$

$\zeta = \frac{9}{6} = 1.5 > 1$ (overdamped)

$\rightarrow G_2(s) = \dfrac{9}{s^2+2s+9}$

$\omega_n^2 = 9 \Rightarrow \omega_n = 3$

$2\zeta\omega_n = 2 \Rightarrow \zeta = \dfrac{2}{6} = \dfrac{1}{3}$

$0 < \zeta < 1$ (underdamped)

$c(t) = 1 - 1.06e^{-t}\cos(\sqrt{8}t - 19.47)$

$\rightarrow G_3(s) = \dfrac{9}{s^2+9}$

$\omega_n = 3$; $2\zeta\omega_n = 0$; $\zeta = 0$ (undamped)

$c(t) = 1 - \cos 3t$

$\rightarrow G_4(s) = \dfrac{9}{(s+3)^2} = \dfrac{9}{s^2+6s+9}$

$2\zeta\omega_n = 6$; $\zeta = 1$; $\omega_n = 3$

(critically damped)

$c(t) = 1 - 3te^{-3t} - e^{-3t}$

Transient Response of Underdamped 2nd order System

1- Rise time:

Time to reach 10-90% of final value.

2- Peak Time:

Time to reach 1st peak of the response

3- Maximum Overshoot

The amount that the waveform overshoots the steady state or final value at the peak time expressed as a percentage of steady state value

$$\%OS = \frac{C_{max} - C_{final}}{C_{final}} \times 100\%$$

4- Settling time

The time constants they take to reach steady state.

Response of second order system

$$C(s) = \frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)} = \frac{A}{s} + \frac{Bs+C}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$A = 1$; $B = -1$; $C = -2\zeta\omega_n$

$$C(s) = \frac{1}{s} - \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n+\omega_n^2+\omega_n^2\zeta^2-\omega_n^2\zeta^2}$$

$$= \frac{1}{s} - \frac{s+2\zeta\omega_n}{(s+\zeta\omega_n)^2+\omega_n^2(1-\zeta^2)}$$

$\therefore \omega_n^2(1-\zeta^2) = \omega_t^2$

$\omega_n\sqrt{(1-\zeta^2)} = \omega_t$

$$= \frac{1}{s} - \left[\frac{s+\zeta\omega_n}{(s+\zeta\omega_n)^2+(\omega_n\sqrt{1-\zeta^2})^2} + \frac{\zeta\omega_n(\sqrt{1-\zeta^2})}{\sqrt{(1-\zeta^2)}\left[(s+\zeta\omega_n)^2+(\omega_n\sqrt{1-\zeta^2})^2\right]}\right]$$

$$c(t) = 1 - \left[e^{-\zeta\omega_n t}\cos(\omega_n\sqrt{1-\zeta^2})t + \frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n(\sqrt{1-\zeta^2})t\right]$$

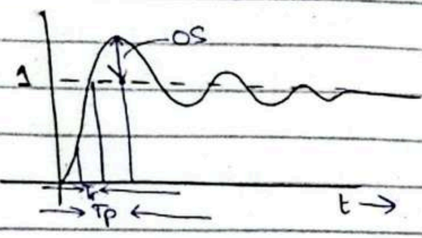
$$= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[\sqrt{1-\zeta^2}\cos(\omega_n\sqrt{1-\zeta^2})t + \zeta\sin(\omega_n\sqrt{1-\zeta^2})t\right]$$

$\therefore \cos A\cos B + \sin A\sin B = \cos(A-B)$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left[\cos(\omega_n(\sqrt{1-\zeta^2})t - \phi\right]$$

$$\boxed{\phi = \tan^{-1}\frac{\zeta}{\sqrt{1-\zeta^2}}}$$ → only remember this

Now will use this expression to find peak time and overshoot-



Peak Time T_p

$$\boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}$$

we can also derive this formula but I'm not gonna do it here.

Max Overshoot

$$c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos\omega_n\sqrt{1-\zeta^2}\,t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_n\sqrt{1-\zeta^2}\,t\right)$$

$C_{max} = C(T_p) \quad \therefore T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

$$c(t) = 1 - e^{-\frac{\zeta\omega_n\pi}{\omega_n\sqrt{1-\zeta^2}}}\left(\cos\omega_n\sqrt{1-\zeta^2}\cdot\frac{\pi}{\cos\omega_n\sqrt{1-\zeta^2}} + \frac{\zeta}{\sqrt{1-\zeta^2}}\left(\sin\omega_n\sqrt{1-\zeta^2}\cdot\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}\right)\right)$$

$$C_{max} = 1 + e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$\boxed{\%OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100}$$ → remember this only

Location of poles of 2nd order system in terms of ζ and ω_n

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = \frac{-b \pm\sqrt{b^2-4ac}}{2a}$$

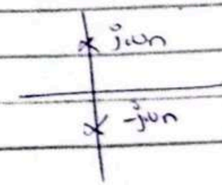
$$s = \frac{-2\zeta\omega_n \pm\sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$= \frac{-\cancel{2}\zeta\omega_n \pm 2\omega_n\sqrt{\zeta^2-1}}{\cancel{2}}$$

$$= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$$

↳ Undamped Condition

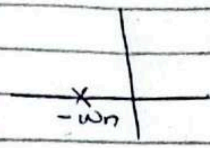
$\zeta = 0 \quad ; \quad s = \pm j\omega_n$



↳ Critically damped condition

$\zeta = 1$

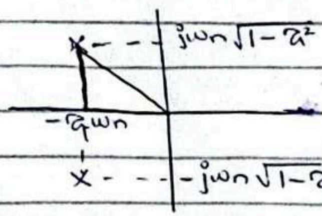
$s = -\omega_n \pm 0$



↳ Underdamped condition

$0 < \zeta < 1$

$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$



$$r = \text{root} = \sqrt{\zeta^2\omega_n^2 + \left(\omega_n\sqrt{1-\zeta^2}\right)^2}$$
$$= \sqrt{\zeta^2\omega_n^2 + \omega_n^2(1-\zeta^2)}$$
$$= \sqrt{\zeta^2\omega_n^2 + \omega_n^2 - \omega_n^2\zeta^2} = \omega_n$$

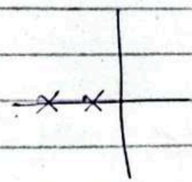
$$\cos\theta = \frac{\zeta\cancel{\omega_n}}{\cancel{\omega_n}} \Rightarrow \theta = \cos^{-1}(\zeta)$$

↳ Overdamped condition

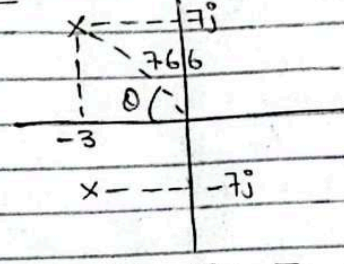
$\zeta > 1$

$s = -\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}$

$s = -\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}$



Question



Find $\zeta, \omega_n, T_p, \%OS, T_s = ?$

$\theta = \tan^{-1}\frac{7}{3}$; $\omega_n = \sqrt{7^2+3^2}$; $\zeta = \cos\theta$

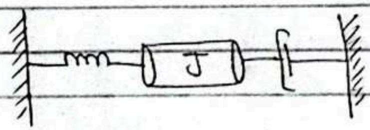
$\omega_n = \sqrt{49+9} = \sqrt{58} = 7.616$; $\zeta = 0.394$

$\theta = 66.80°$

$T_p = \frac{\pi}{7}$; $T_s = \frac{4}{\zeta\omega_n}$; $OS = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$

$T_p = 0.449\,sec$; $T_s = 1.33\,sec$; $OS = 26\%$

Question



$K = 5\ Nm/rad$

Find J and D to yeild 20% OS and $T_s = 2\,sec$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d} \quad ; \quad T_s = \frac{4}{\zeta\omega_n}$$

$$\%OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$\theta s(Js^2 + Ds + K) = Ts$$

$$\frac{\theta s}{Ts} = \frac{1}{Js^2 + Ds + K}$$

$$\frac{J}{J} \times \frac{1}{Js^2 + Ds + K} = \frac{1/J}{s^2 + \frac{D}{J}s + \frac{K}{J}}$$

$$\omega_n^2 = \frac{K}{J} \quad ; \quad 2\zeta\omega_n s = \frac{Ds}{J}$$

$$\omega_n = \sqrt{\frac{K}{J}} \quad ; \quad \zeta = \frac{D}{J} \cdot \frac{1}{2\omega_n}$$

$$0.2 = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100$$

$$\zeta = \frac{-\ln(0.2/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$20\%\ OS \rightarrow \zeta = 0.456$

$T_s = 2\,sec$

$$T_s = \frac{4}{\zeta\omega_n} \Rightarrow \zeta\omega_n = \frac{4}{2} \Rightarrow \omega_n = \frac{2}{\zeta} = 4.4$$

Chapter # 4

Relationship b/w 'ζ' and 'overshoot'

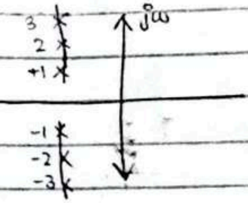
* Everytime corresponds to ζ and overshoot

→ inverse relation (b/w ζ and overshoot)

overshoot ↑ ζ ↓

→ direct relation (b/w rise time and ζ)

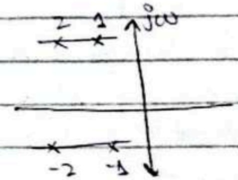
ζ ↑ rise time ↑

1) 

$(\text{peak time})_{max} = 1$; $(\text{peak time})_{min} = 3$

$(\text{overshoot})_{max} = 3$; $(\text{overshoot})_{min} = 1$

∴ all poles on vertical lines have same 'settling time'

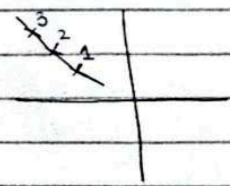
2) 

peak time → same

$(\text{settling time})_{max} = 1$

$(\text{overshoot})_{max} = 1$

∴ poles closer to origin have high settling time

3) 

(ζ and OS) → same

(settling & T_p) → different

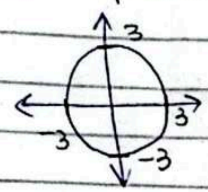
$(\text{settling time})_{max} = 1$

$(\text{peak time})_{max} = 3$

* every real pole produces an exponential
* if pole is far from origin, we neglect it

S-Plane

- Circle of 3 in s-plane

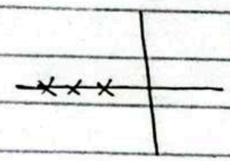


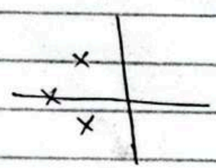
Question

if $\zeta = 0.5$ & $\omega_n = 2$
Find transfer fn

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} = \frac{4}{s^2 + 2s + 4}$$

⊙ if peaks are same → ζ and OS are same
* small ω_n → less oscillations
* high ω_n → more oscillations

-  ⇒ combination of three first order system

-  ⇒ pole – zero plot of third order system

Chapter # 6

Stablity

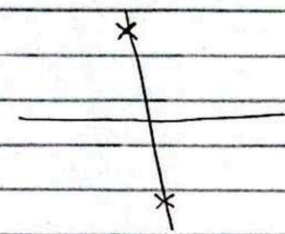
Requirements of the design of control systems

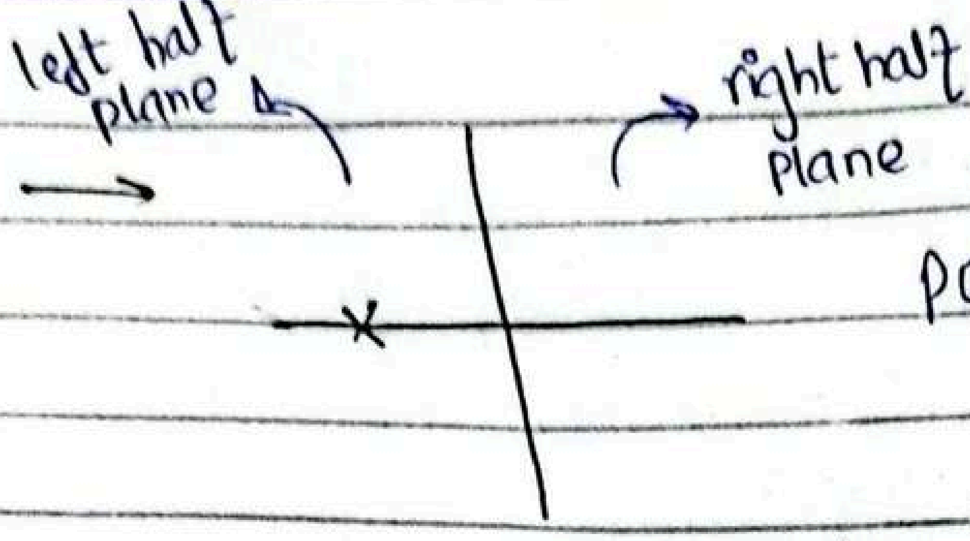
- Transient Response
- Stability
- Steady state error

* System should be stable

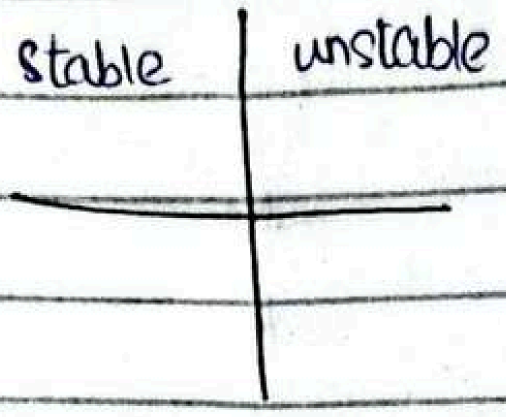
- BIBO → stability in terms of total response
- BIUO → unstable

→ using natural response
- if 't' in natural response approaches zero, it is a stable system
- if 't' approaches ∞ – unstable system
- if natural response neither decays nor grows but remain constant / oscillates, so it is a marginally stable system

→ 
oscillatory response
(Marginally stable system)



Pole of multiplicity 'ONE'



SUFFICIENT CONDITIONS FOR INSTABILITY

- Signs of denominator's coeffecient are not same

  Example:

  $$\frac{3}{s^3-2s^2+5s+3}$$

  $(+1, -2, +5, +3) \rightarrow$ not same

  $\Rightarrow 1s^3-2s^2+5s+3=0 \rightarrow$ characteristics Equation

- If any power of 's' is missing

  Example:

  $$\frac{3}{s^3-2s+3}$$

  s^2 is missing